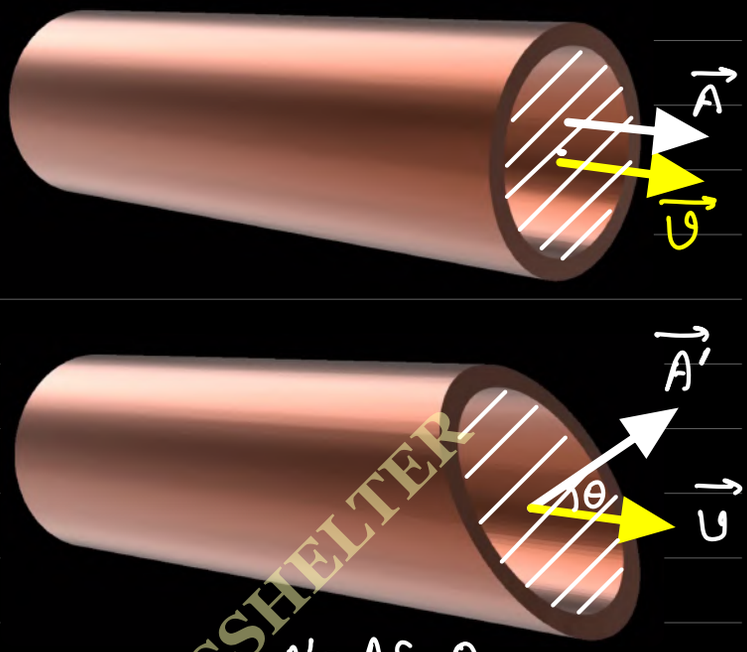
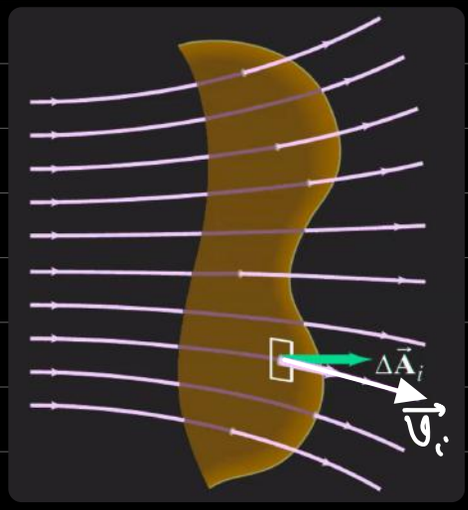
FLUX : Flux refers to flow (for instance the word "efflux" means outflow). In the content of fluid mechanics, flux refers to volumetric flow rate. Suppose a fluid were to flow through a pipe of cross sectional Area A with a velocity v then if Area is perpendicular to velocity then we know from our understanding of fluid mechanics that flow rate is $\phi = Av$. Now if we cut the pipe mouth by an oblique plane as shown in the figure, the area of the outlet increases by a factor of $\sec\theta$, however flow rate still remains the same. If A' be the new area of the oblique outlet face then we can write $\phi = A'v\cos\theta$. Now looking at $\cos\theta$ gives us the exciting possibility of writing the flux or the flow rate as dot product of "Area vector" and the velocity vector. Here we define Area Vector as a vector whose magnitude is equal to the area of the oblique face and direction being $\perp$ to the oblique face. Thus we can write the Volumetric flux (or flow rate) as:



$$\boxed{\phi = \vec{A'} \cdot \vec{v}}$$

Expressing the flow rate can be very useful mathematically for calculating flow rates through arbitrary surfaces. For instance if we were to tie a porous cloth on the mouth of the pipe then if $\vec{v_i}$ is the velocity vector associated with a patch of area $\Delta\vec{A_i}$ then the flow rate through the patch will be $\vec{v_i} \cdot \Delta\vec{A_i}$ and net flow rate will be the summation: $\phi = \sum \vec{v_i} \cdot \Delta\vec{A_i}$



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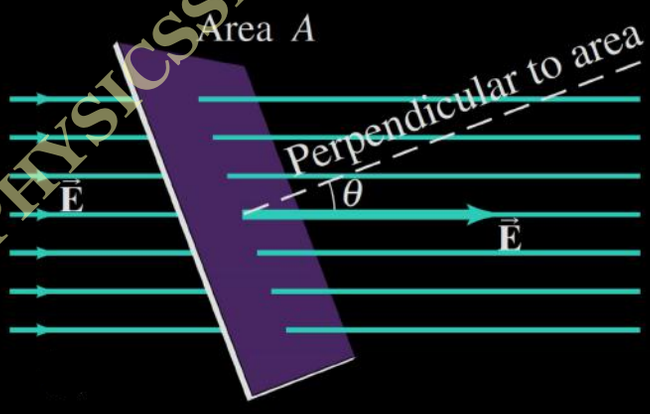
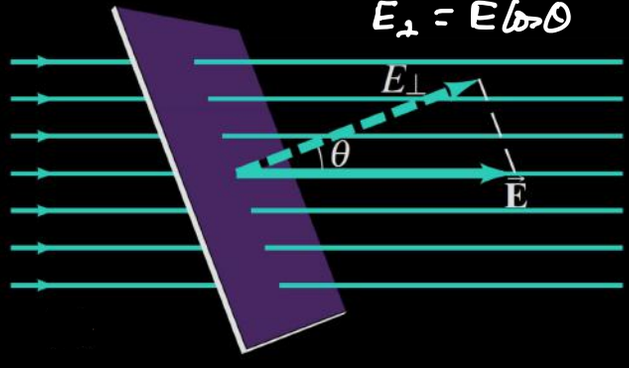
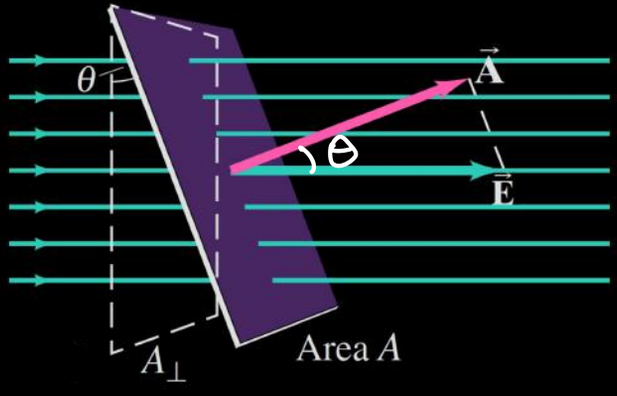
Instead of using summation of notation we could alternatively write volume flow rate though the surface as:

$\phi = \int \vec{v} \cdot d\vec{A}$ where $\vec{v}$ is the velocity vector associated with a patch of area $d\vec{A}$. The ideas similar to flow of water though patches is useful for visualising electric field line patterns associated with patches of areas (imagined or real) in space and we will be able to appreciate the beauty of this visualisation when we will be able to use it for evaluating electric fields due to various charge geometries.

ELECTRIC FLUX: Parallel to the idea of fluid flux, we define the Electric flux. If $\vec{E}$ is the electric field passing an Area A whose $\perp$ makes an angle θ with the local Electric field, then the Electric flux associated with the area is given as $\phi = EA\cos\theta$. If $E_\perp$ were component of electric field perpendicular to this area, we could also write the electric flux as $\phi = E_\perp A$. (See adjacent figure). Also instead of resolving the electric field, we could also consider projection of the area $\perp$ to the electric field. As clear from adjacent figure, $A_\perp = A\cos\theta$, thus we could also write $\phi = EA_\perp$. Now if we were to define Area Vector having a direction $\perp$ to the plane of the surface and magnitude equal to







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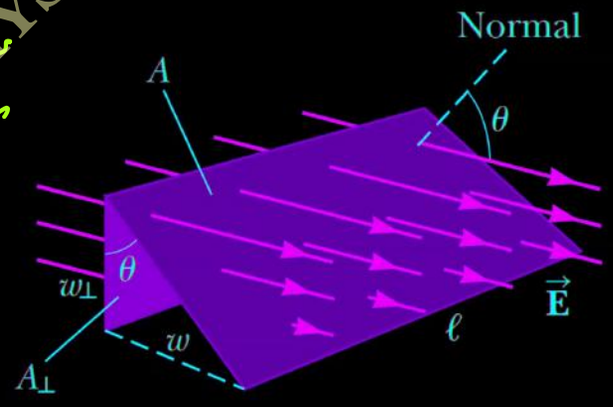
the area of the surface then we can readily see that we can write flux as $\phi = \vec{E} \cdot \vec{A}$.

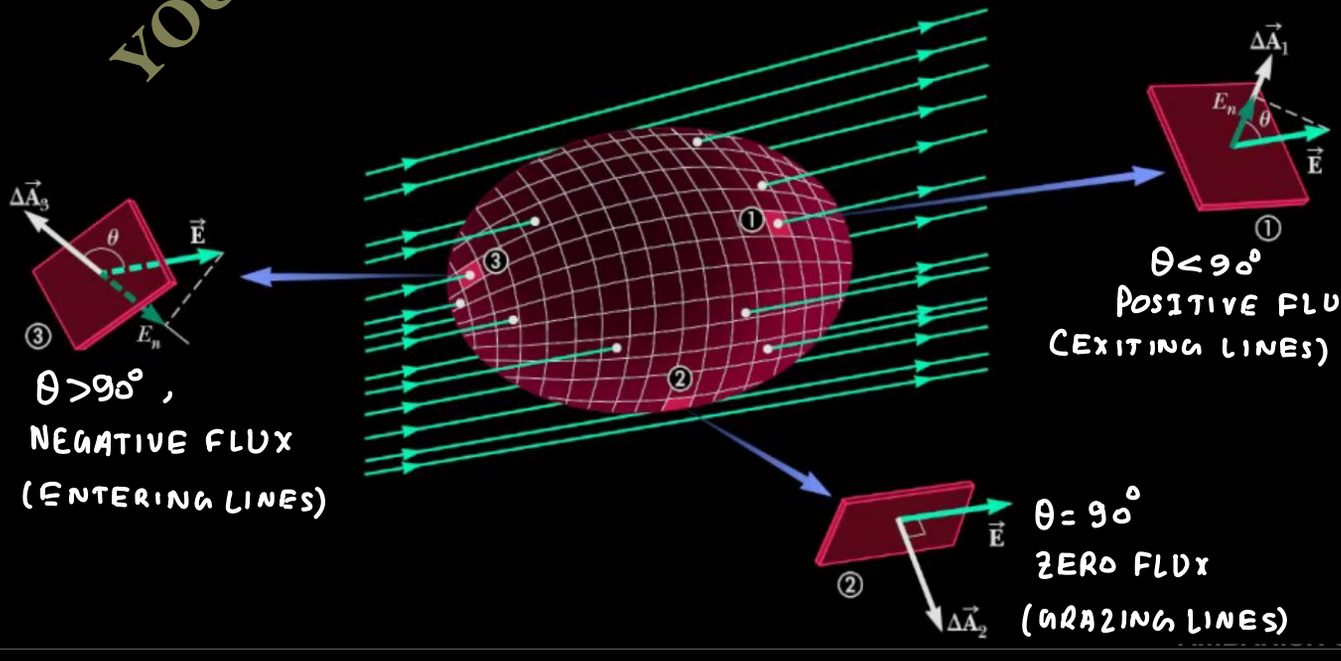
→ Note that there exist two normals to any surface and we could choose any of the normals, there is no particular convention regarding this, however choosing the other normal would yield an exact negative of the calculated flux. The sign of flux simply tells us that whether "flow" is making an acute angle or obtuse angle with chosen normal. A positive flux means the flux crossing towards the normal side of the surface and a negative flux means that flow is crossing against the normal side of the surface.

→ When we have a choice, I recommend choosing a normal which makes an acute angle with field, this gives a positive flux and I am a positive minded person ☺.

→ Note that same number of field lines pass through the $A_\perp$ as through the given surface and flux is just a measure of number of field lines crossing an area. If same number of field lines cross though two surfaces, then their fluxes are equal.



→ For Closed Surfaces Normal is always defined as outward normal.



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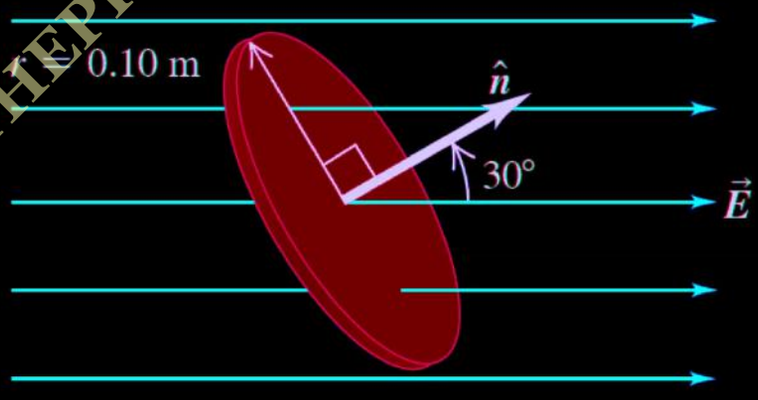
By this convention it can be easily seen that entering field lines in a closed surface will create a negative flux ($\because \theta > 90°$) and exiting field lines will create positive flux ($\theta < 90°$), while field lines parallel to any patch do not create any flux.

EXAMPLE: YOUNG & FREEDMAN:

EXAMPLE 22.1 Electric flux through a disk

A disk of radius 0.10 m is oriented with its normal unit vector $\hat{n}$ at 30° to a uniform electric field $\vec{E}$ of magnitude 2.0×10^3 N/C (**Fig. 22.7**). (Since this isn't a closed surface, it has no "inside" or "outside." That's why we have to specify the direction of $\hat{n}$ in the figure.) (a) What is the electric flux through the disk? (b) What is the flux through the disk if it is turned so that $\hat{n}$ is perpendicular to $\vec{E}$? (c) What is the flux through the disk if $\hat{n}$ is parallel to $\vec{E}$?



Solution: (a) $\phi = \vec{E} \cdot \vec{A} = EA\cos\theta$

$= 2.0 \times 10^3 \times \pi (0.10)^2 \times \cos 30°$

$\Rightarrow \phi = 54\ \text{N m}^2/\text{C}$

(b) If $\hat{n} \perp \vec{E}$ then $\vec{E} \cdot A\hat{n} = 0$

$\Rightarrow \phi = 0$

(c) If $\hat{n} \parallel \vec{E} \Rightarrow \vec{E} \cdot A\hat{n} = EA$

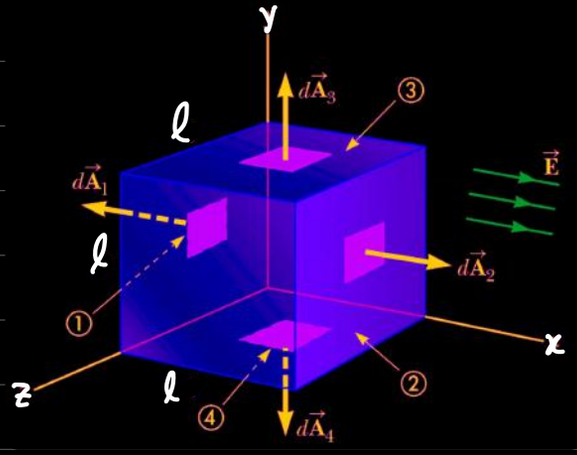
$2.0 \times 10^3 \times \pi (0.1)^2 \times \cos 0° = 63\ \text{N m}^2/\text{C}$. Ans.

EXAMPLE: FLUX THROUGH A CUBE (UNIFORM FIELD):

Consider a cube of edge ℓ placed in a uniform electric field $\vec{E} = E\hat{i}$. Find the flux through all faces of the cube and hence find the net flux through the cube.

Solution: Since field lines are parallel to all faces of the cube except ① & ② ($\perp$ to x axis), $\therefore$ flux through all other faces is zero. Now for face ①:

$\phi_1 = E \times \ell^2 \times \cos 180° = -E\ell^2$



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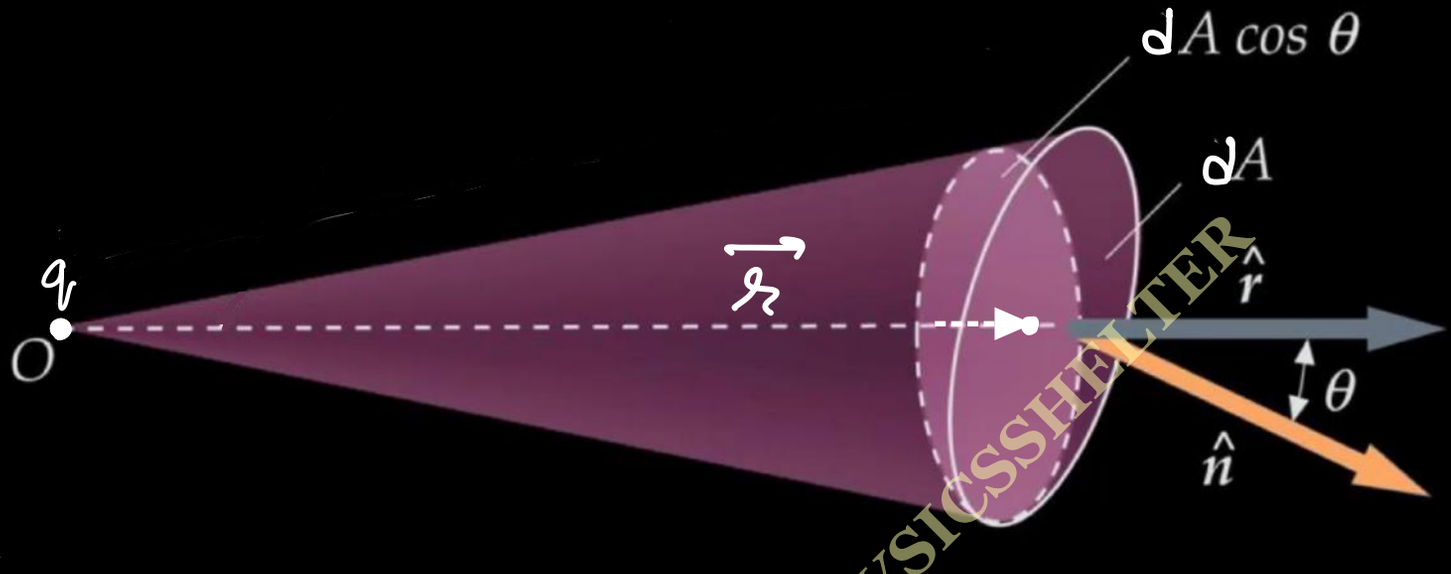
Similarly for face ②, $\phi_2 = E\ell^2 \cos 0^\circ = E\ell^2$. Now net flux

$\phi = \phi_1 + \phi_2 = E\ell^2 + (-E\ell^2) = 0$ Ans.

Flux Per Unit Solid Angle Due To A Point Charge:



Consider an elemental patch of area dA at a distance r from a point charge q. Due to this point charge, field at the patch is given as:

$$\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{r}$$, Now flux through the patch,

$$d\phi = \vec{E}\cdot\vec{dA} = \frac{1}{4\pi\epsilon_0}\cdot\frac{q}{r^2}\cdot dA\cos\theta = \frac{q}{4\pi\epsilon_0}\left(\frac{dA\cos\theta}{r^2}\right)$$

Now recall that $(dA\cos\theta/r^2)$ represents elemental solid angle subtended by the patch at the point charge. Thus, we can write: $d\phi = \frac{q}{4\pi\epsilon_0}d\omega$ where $d\omega = \frac{dA\cos\theta}{r^2}$ (i.e solid angle).

$$\Rightarrow \boxed{\frac{d\phi}{d\omega} = \frac{q}{4\pi\epsilon_0}}$$

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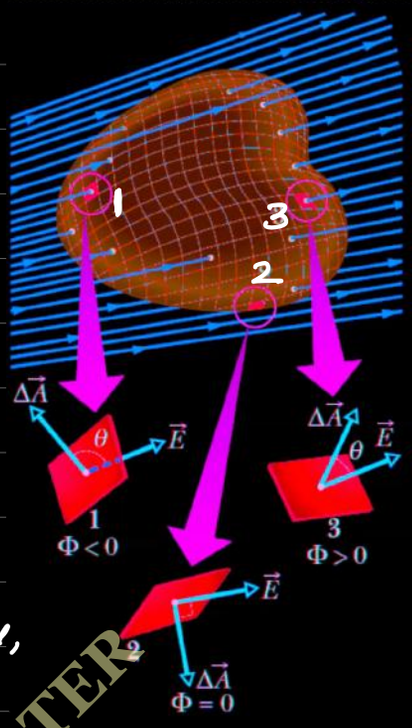
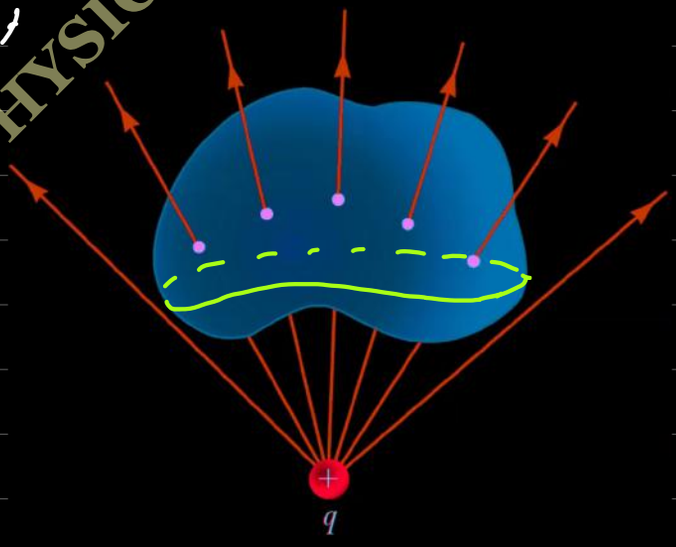
NEGATIVE SOLID ANGLE: Once we are defining flux per unit solid angle, how do we account for negative flux associated with some patches. To overcome this problem, we can assign a sign to the solid angle. i.e we shall consider solid angle to be positive if the angle between $\hat{n}$ and $\vec{A}$ is acute and negative if angle between $\hat{n}$ and $\vec{A}$ is obtuse. Thus if for the adjacent figure if we imagine field lines to be diverging from a point charge outside the closed surface, then solid angle for patch ① is negative while for patch ③ it is positive.



FLUX DUE TO A POINT CHARGE OUTSIDE A CLOSED SURFACE:

It can be readily seen that if a point charge lies outside a closed surface, then because of the point charge, the number of field lines entering the close surface is equal to the number of field lines leaving and hence due to an outside point, flux through a closed surface is zero.



We can also think about it in terms of solid angles. Consider infinite number of rays emanating from the point charge, due to this certain region of the closed surface will be illuminated and a part of the surface will be in darkness. Both the surfaces can be said to be mounted on the same perimeter (i.e the contour separating the dark portion from the illuminated one). Now being mounted on the same contour, the magnitude of solid angles by both parts is same however the solid angle of illuminated portion is negative and that of dark portion is positive.

→ **THUS FLUX THROUGH A CLOSED SURFACE DUE AN OUTSIDE CHARGE IS ALWAYS ZERO.**

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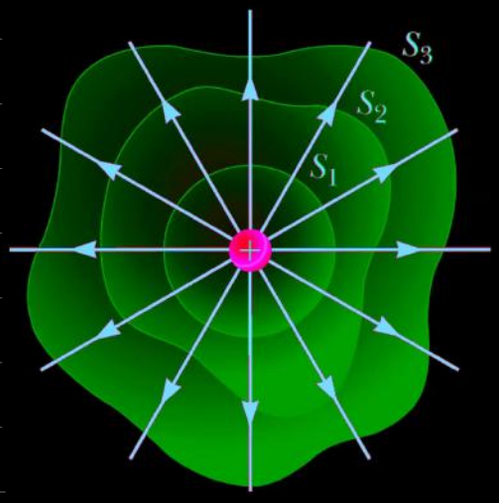
FLUX DUE TO A POINT CHARGE INSIDE A CLOSED SURFACE:



Recall that solid angle subtended by any closed surface at an inside point is 4π steradians. Also recall that flux per unit solid angle due to a point charge is given as:

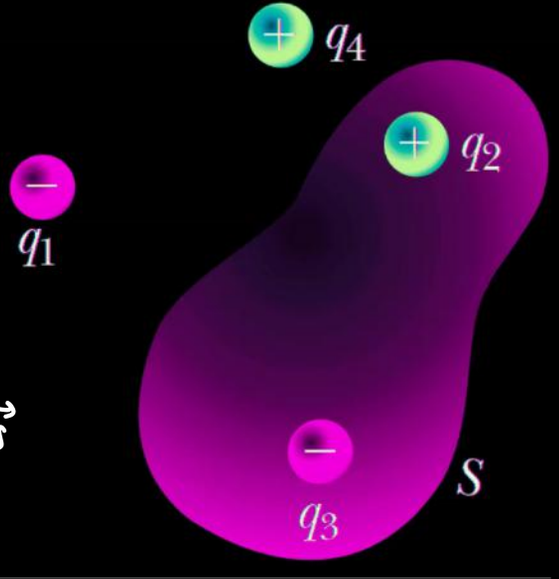
$$\frac{d\phi}{d\Omega} = \frac{q}{4\pi\epsilon_0} \Rightarrow \phi = \frac{q}{4\pi\epsilon_0}\Omega$$

Thus for a completely surrounding surface,

$$\phi = \frac{q}{4\pi\epsilon_0} \times 4\pi = q/\epsilon_0$$

i.e $\oint \vec{E}\cdot d\vec{S} = q/\epsilon_0$, Here a circle on the integral sign means that integration is being done over the complete closed surface. Also note that exact shape of the surface does not matter, thus flux through S_1, S_2 and S_3 each is q/ϵ_0 ∵ all of them surround the charge completely.

FLUX THROUGH A CLOSED SURFACE DUE TO SEVERAL CHARGES:



Consider a closed surface S along with several other charges in the universe for instance q_1, q_2, q_3 and q_4. Some of them inside and some of them outside S. Then, net flux through S is given as:

$$\phi = \oint \vec{E}\cdot d\vec{S} = \oint (\vec{E}_1 + \vec{E}_2 + \vec{E}_3 + \vec{E}_4)\cdot d\vec{S}$$

By distributive property of dot product, we can write $\phi = \oint \vec{E}_1\cdot d\vec{S} + \oint \vec{E}_2\cdot d\vec{S} + \oint \vec{E}_3\cdot d\vec{S} + \oint \vec{E}_4\cdot d\vec{S}$,

which simply a superposition of individual fluxes due to each charge. Now recall that outside charges produce zero flux, while the inside charges produce a flux of q/ϵ_0, thus in this

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particular case, $\phi = 0 + \frac{q_2}{\epsilon_0} + \frac{q_3}{\epsilon_0} + 0$ i.e $(q_2+q_3)/\epsilon_0$

this understanding leads us to the formal statement of the Gauss law.

GAUSS LAW: Net Electric flux associated with any closed surface is equal to $1/\epsilon_0$ the net charge enclosed by it. Mathematically,

$$\oint \vec{E} \cdot d\vec{S} = \frac{q_{in}}{\epsilon_0}$$

Although Gauss law is always applicable, it is usually beneficial when we can find a "Gaussian" surface which is everywhere parallel or perpendicular to field lines. Usually we shall use Gauss law for three kinds of symmetries namely planar symmetry (infinite sheets or slabs with charge density being same in a plane), cylindrical symmetry (e.g. infinite line charges, solid cylinders and cylindrical shells with same charge density at a given radius) and spherical symmetry (i.e. spherical shells or solid spheres with charge densities varying only as a function of radius).

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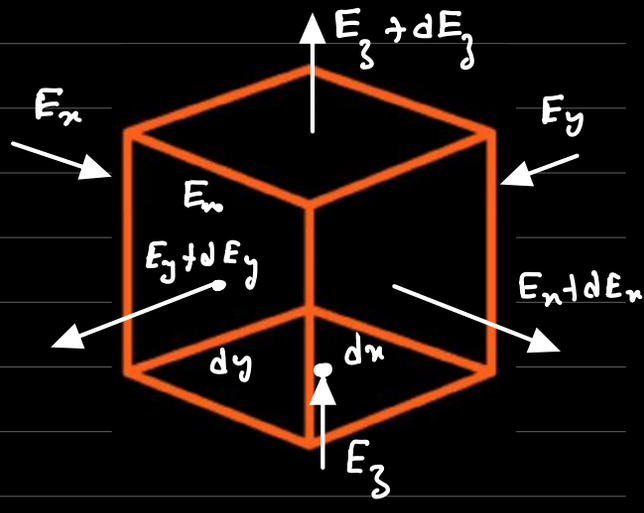
DIFFERENTIAL FORM OF GAUSS LAW (CARTESIAN COORDINATES):

Consider a Gaussian cube in space with sides dx, dy and dz and electric fields at planar faces with coordinate x, $x+dx$, y, $y+dy$, z and $z+dz$ as shown in the figure. Let ρ be the local charge density inside the cube. Now applying Gauss law to the cube,



$$(E_x + dE_x)\,dy\,dz - E_x\,dy\,dz + (E_y + dE_y)(dz\,dx) - E_y\,dz\,dx + (E_z + dE_z)\,dx\,dy - E_z\,(dx\,dy) = \rho\,dx\,dy\,dz/\epsilon_0.$$

$\Rightarrow dE_x\,dy\,dz + dE_y\,dx\,dz + dE_z\,dx\,dy = \rho\,dx\,dy\,dz/\varepsilon_0$

Dividing throughout by $dx\,dy\,dz$:

$$\frac{dE_x}{dx} + \frac{dE_y}{dy} + \frac{dE_z}{dz} = \frac{\rho}{\varepsilon_0},$$ However since E_x, E_y and E_z are all multivariable functions, here by dE_x/dx etc we actually mean partial derivatives. Thus we have.

$$\boxed{\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} = \frac{\rho}{\varepsilon_0}} \quad \text{or} \quad \boxed{\vec{\nabla}\cdot\vec{E} = \rho/\varepsilon_0} \quad (1)$$

Equation (1) is known as Poisson's equation and when in free space, $\rho = 0$ and the equation becomes:

$\vec{\nabla}\cdot\vec{E} = 0$, this is known as Laplace's Equation.

the term $\vec{\nabla}\cdot\vec{E}$ is read as divergence of $\vec{E}$ because in a sense it is a measure of how many field lines are emerging out of a given a point in space. (I would have preferred to term it as "emergence"). Also recall that $\vec{E} = -\nabla V$, thus we could also write differential form of Gauss law as:

$$\frac{\partial}{\partial x}\left(-\frac{\partial V}{\partial x}\right) + \frac{\partial}{\partial y}\left(-\frac{\partial V}{\partial y}\right) + \frac{\partial}{\partial z}\left(-\frac{\partial V}{\partial z}\right) = \rho/\varepsilon_0$$

or $$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = -\rho/\varepsilon_0 \Rightarrow \nabla^2 V = -\rho/\varepsilon_0$$

where ∇^2 is the operator $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)$ also known as "Laplacian" operator.

EXAMPLE : Field in certain region of space is given as: $\vec{E} = \frac{x\hat{i} + y\hat{j} + z\hat{k}}{x^2+y^2+z^2}$. Compute charge density in space, also calculate flux through a sphere of radius R centred at the origin.

Solution: $$\frac{\partial E_x}{\partial x} = \frac{(x^2+y^2+z^2)\times 1 - x\times 2x}{(x^2+y^2+z^2)^2} = \frac{-x^2+y^2+z^2}{(x^2+y^2+z^2)^2}$$

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Similarly We can find $\partial E_y/\partial y$ and $\partial E_z/\partial z$ by cyclic substitution of $x \to y \to z \to x$. Now, by divergence theorem:

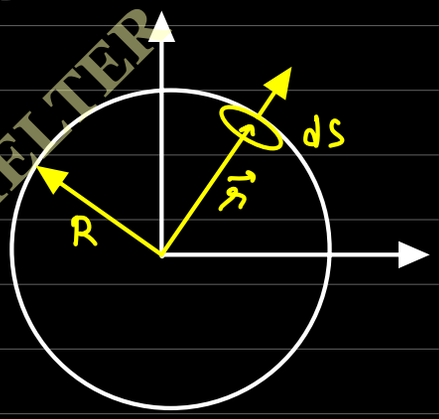
$$\rho = \frac{x^2+y^2+z^2}{(x^2+y^2+z^2)^2} \times \epsilon_0 = \frac{1}{r^2}\epsilon_0$$

Now Total charge enclosed in radius R, $Q = \int_0^R \rho \times 4\pi r^2 dr = 4\pi\epsilon_0 R$ Ans.

$\phi = 4\pi R$

Alternatin Solution: $\vec{E} = \frac{xi+yj+zk}{x^2+y^2+z^2} = \frac{\vec{r}}{r^2}$

For surface of our sphere, $r = R$. We shall try to apply gauss law for our spherical surface. Consider a patch of area $d\vec{s}$ at position vector $R\hat{r}$. Now $d\phi = \vec{E}\cdot d\vec{s}$ $\Rightarrow$



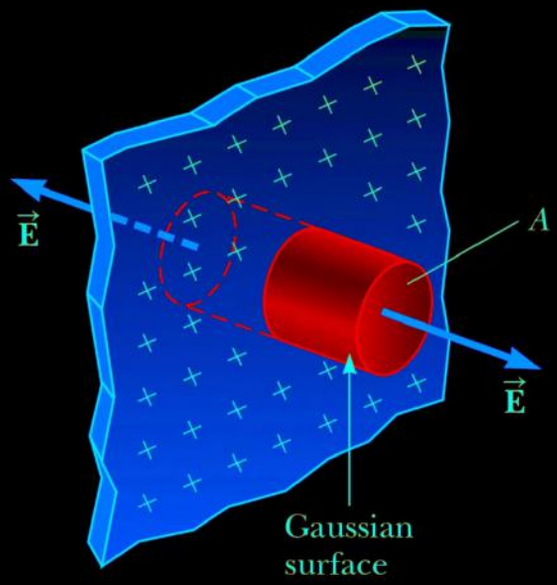
$$d\phi = \frac{R\hat{r}}{R^2}\cdot(ds\,\hat{r}) = \frac{ds}{R}$$

$$\Rightarrow \phi = \frac{S}{R} = \frac{4\pi R^2}{R} = \frac{Q_{enc}}{\epsilon_0} \Rightarrow Q_{enc} = 4\pi R\epsilon_0$$

$\phi = Q_{enc}/\epsilon_0 = 4\pi R$ Ans.

Field Due To Infinite Sheet Of Charge:

Recall that we had earlier found the field due to infinite sheet of charge using the formula for normal field $E_N = k\sigma\omega$. (Also using formula for field due to disc putting $R \to \infty$). Now we can find the same using Gauss Law. Imagine a cylinder shape gaussian surface with cross section area as A. Flux through curved surfaces of the Gaussian Cylinder Must be zero ($\because \vec{E} \parallel \vec{S}$), Now Flux through flat lids of the cylinder:



$$\phi = E \times A \times 2 = (\sigma \times A)/\epsilon_0 \quad (\text{Gauss Law}, Q_{enc}/\epsilon_0)$$

Thus $\boxed{E = \frac{\sigma}{2\epsilon_0}}$

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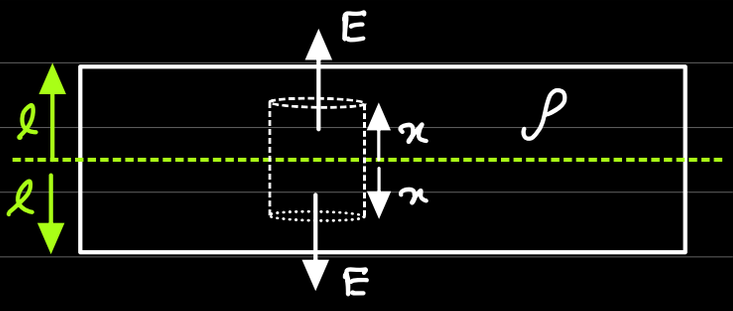
ELECTRIC FIELD DUE TO AN INFINITE THICK SHEET :

Consider an infinitely wide slab of uniform charge density having thickness $2l$. We wish to find field inside and outside the sheet. First consider the Gaussian cylinder of height $2x$ where $x < l$. Let field here be E and A be cross section of our Gaussian cylinder then by Gauss law:

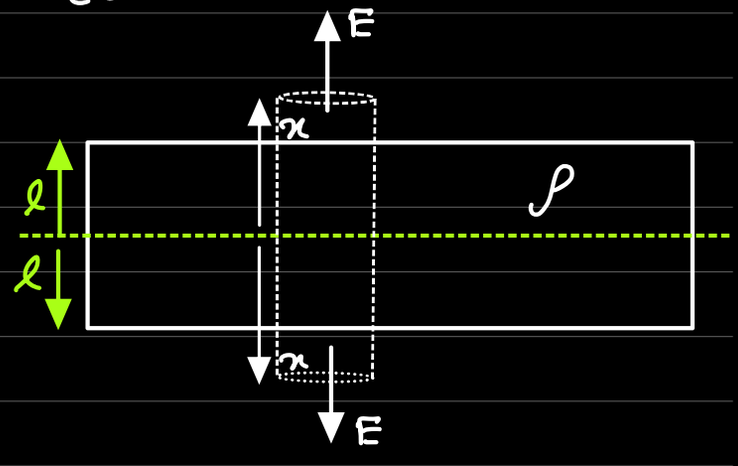


$$2EA = (\rho \times 2Ax)/\epsilon_0 \Rightarrow \boxed{E = \rho x/\epsilon_0}$$

Thus field at $x = l$ = $E = \rho l/\epsilon_0$ and for outside points, each layer of the slab serves as infinite sheet and ∴ creates a constant field. The charge per unit area of the sheet is given by:

$$\sigma_{eff} = \left(\frac{\rho \times A \times 2l}{A}\right) = 2\rho l$$ and ∴ field outside the sheet is given as $E = \frac{\sigma_{eff}}{2\epsilon_0} = \frac{2\rho l}{2\epsilon_0} = \frac{\rho l}{\epsilon_0}$.

Alternatively we could also use Gauss law as shown in the figure:

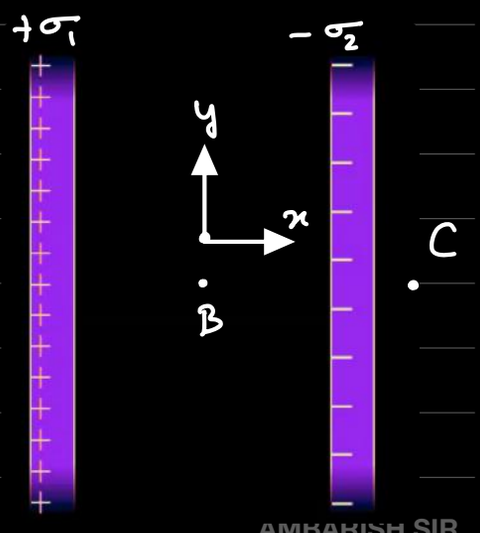


$2EA = (\rho \times A \times 2l/\epsilon_0)$

$\Rightarrow E = \rho l/\epsilon_0$ Ans.

EXAMPLE Two infinite parallel plates are arranged as shown in the figure, find:

(a) Field in regions A, B & C (b) Force of attraction per unit area between the sheets.



Solution: Using superposition and understanding of field due to infinite sheets:

$$\vec{E}_A = \left(-\frac{\sigma_1}{2\epsilon_0} + \frac{\sigma_2}{2\epsilon_0}\right)\hat{i} = (\sigma_2 - \sigma_1)/2\epsilon_0\ \hat{i}$$

$$\vec{E}_B = \left(\frac{\sigma_1}{2\epsilon_0} + \frac{\sigma_2}{2\epsilon_0}\right)\hat{i}, \quad \vec{E}_C = \left(\frac{\sigma_1}{2\epsilon_0} - \frac{\sigma_2}{2\epsilon_0}\right)\hat{i}$$

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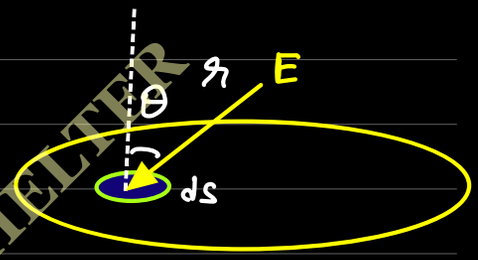
For finding force on any sheet we must consider only field created by the other sheet. Now field at negative sheet due to positive sheet, $\vec{E}_+ = \frac{\sigma_1}{2\epsilon_0}\hat{i}$, $\vec{F} = \vec{E}q \Rightarrow \vec{F} = \frac{\sigma_1}{2\epsilon_0} \times (-\sigma_2 A)\hat{i}$

$\Rightarrow$ Force per unit area $= \vec{F}/A = -[(\sigma_1\sigma_2)/2\epsilon_0]\hat{i}$ Ans.

Relation Between Normal Force And Flux On A Lamina:

Consider a Patch of area dS on a lamina whose surface charge density is σ. Let local electric field at this lamina due to other charges in universe be $\vec{E}$, then normal component of force on this lamina is given as:



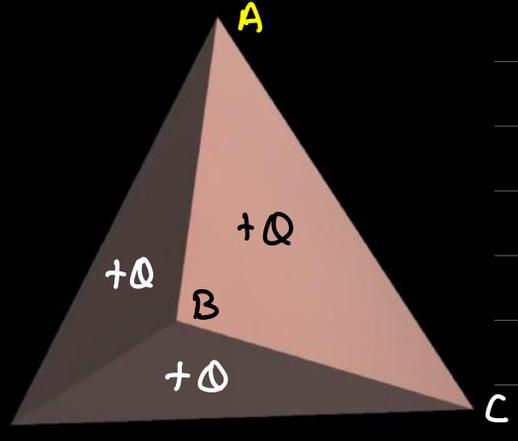
$$dF_N = E \times \sigma dS \cos\theta = \sigma \times d\phi$$

Thus net normal force on the lamina will be given as $F_N = \int \sigma \times d\phi$ and if σ is constant then we can write

$$\boxed{F_N = \sigma \times \phi}$$

→ Note that while calculating flux, only fields due to other charges have to be considered and not the field created by the lamina itself.

Example: Each face of a regular tetrahedron is given a uniformly distributed charge Q. Find the force of electrostatic Repulsion experienced by one of the faces due to other three faces. Area of each face is A.



Solution: Let Area of each face be A. Now imagine a Gaussian surface just outside this tetrahedron (tightly hugging it though). The flux through such a Gaussian surface will be $4Q/\epsilon_0$, and thus flux through one of the faces will be Q/ϵ_0. Imagine this face to be just outside face ABC of the tetrahedron. For this surface, field produced by face ABC everywhere is $\sigma/2\epsilon_0$ i.e $Q/2A\epsilon_0$, $\therefore$ Flux due to other

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three faces must be $(Q/\epsilon_0 - (Q/2A\epsilon_0)A) = \frac{Q}{2\epsilon_0}$.

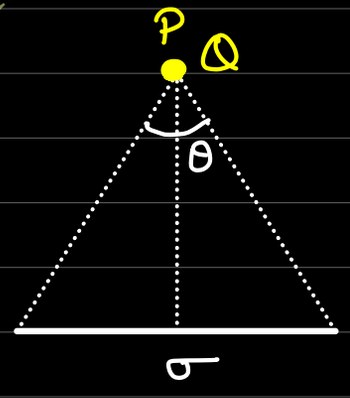
Now Force = $F_N = \sigma \times \phi = \frac{Q}{A} \times \frac{Q}{2\epsilon_0} = \frac{Q^2}{2A\epsilon_0}$ Ans.

Here we can also find average pressure on the face, $p_{avg} = F/A$
thus $p_{avg} = \sigma^2/2\epsilon_0$.

THINK YOURSELF: What would be the average pressure on one of the faces of a cube, hexagonal infinite tube, octahedron, dodecahedron infinite tube given charge per unit area is σ.
Ans $\sigma^2/2\epsilon_0$ (Use exactly the same method as done for tetrahedron).

FIELD ABOVE MIDLINE OF INFINITE STRIP:

Imagine a point charge Q placed at a point P subtending an angle θ on a strip of charge density σ as shown in figure. Now by symmetry, the flux through the infinite strip is given as: $\phi = \frac{\theta}{2\pi}\frac{Q}{\epsilon_0}$,



Thus force on strip, $F_N = \sigma \times \phi = QE_p = \frac{1}{2\pi\epsilon_0}\theta Q\sigma$

$\Rightarrow E_p = \frac{\sigma\theta}{2\pi\epsilon_0}$ Ans.

Alternative Solution: It would take $2\pi/\theta$ number of such strips to form a closed surface around P which would subtend a solid angle of 4π steradians at P. Thus solid angle subtended by the strip : $\omega = 4\pi/(2\pi/\theta) = 2\theta$. Now recall the normal field formula, $E_N = k\sigma\omega = \frac{1}{4\pi\epsilon_0} \times \sigma \times 2\theta = \frac{\sigma\theta}{2\pi\epsilon_0}$ Ans

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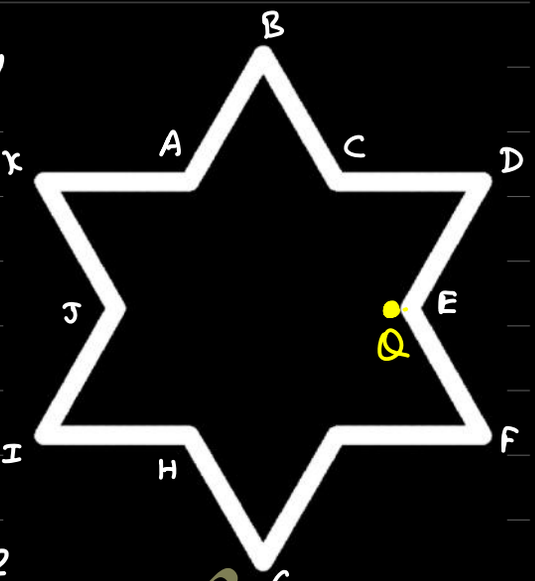
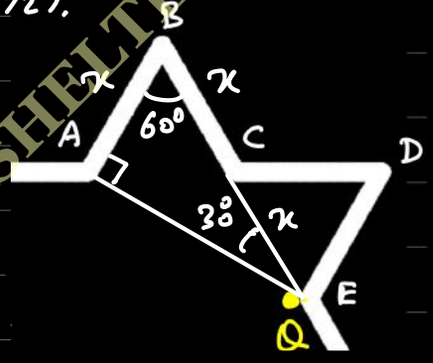
EXAMPLE: An infinite star shaped tube is made by welding several strips as shown, find flux through strip AB due to point charge well within the tube just inside the vertex E as shown. Also find the flux through strip ED



Solution: Consider the construction in adjacent figure. $\because$ ABE = 60°, and AB and BE are in 1:2 ratio $\therefore$ ABE must be a right triangle ($\cos 60° = 1/2$). Thus angle subtended by AB at Q = 30° or $\pi/6$

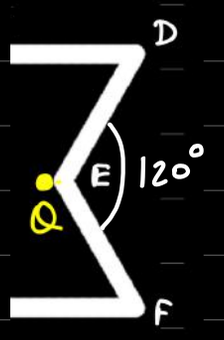
$\therefore$ Flux = $\frac{30°}{360°} \times \frac{Q}{\epsilon_0} = \frac{Q}{12\epsilon_0}$ Ans.



Now refer to the next figure, the total angle subtended by strips ED and EF is 120° $\therefore$ their combined flux must be

$\frac{120°}{360°} \times \frac{Q}{\epsilon_0} = \frac{Q}{3\epsilon_0}$, and due to symmetry this flux is equally distributed through ED and EF

$\therefore$ Flux through ED $\equiv \phi_{FE} = \frac{Q}{6\epsilon_0}$ Ans.



EXAMPLE: PATHFINDER MCQ13 :

A half cylinder of radius R and length $L >> R$ is formed by cutting a cylindrical pipe made of an insulating material along a plane containing its axis. The rectangular base of the half cylinder is closed by a dielectric plate of length of length L and width $2R$. A charge Q on the half cylinder and a charge q on the dielectric plate are uniformly sprinkled. Electrostatic force between the plate and the half cylinder is closest to

(a) $\frac{qQ}{2\varepsilon_0 RL}$ (b) $\frac{qQ}{2\pi\varepsilon_0 RL}$

(c) $\frac{qQ}{4\varepsilon_0 RL}$ (d) $\frac{qQ}{8\varepsilon_0 RL}$

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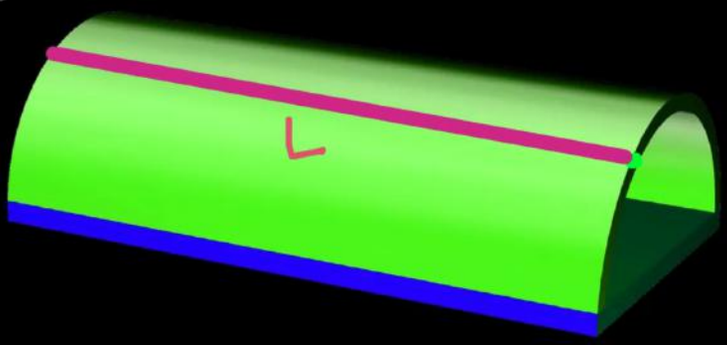
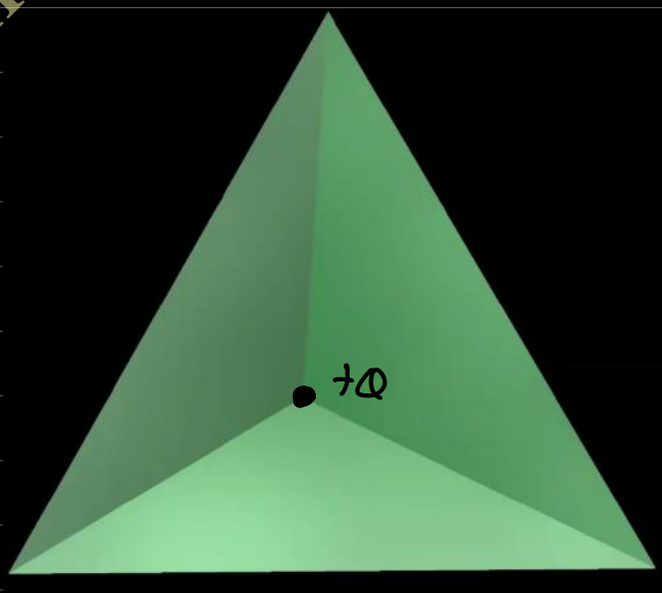
Solution: Consider a strip of charge along the curved semicylinder as shown. This subtends an angle 90° at the base as shown in the next figure. Thus if it has a charge dq, then because of this the flux through dielectric plate is $dq/4\epsilon_0$. Thus if total charge on cylindrical portion is Q, flux through plate is $\phi = Q/4\epsilon_0$. Now $F = \sigma\phi$, Thus force on flat plate is:



$$F = \frac{q}{2RL} \cdot \frac{Q}{4\epsilon_0} = \frac{Qq}{8\epsilon_0 RL}$$

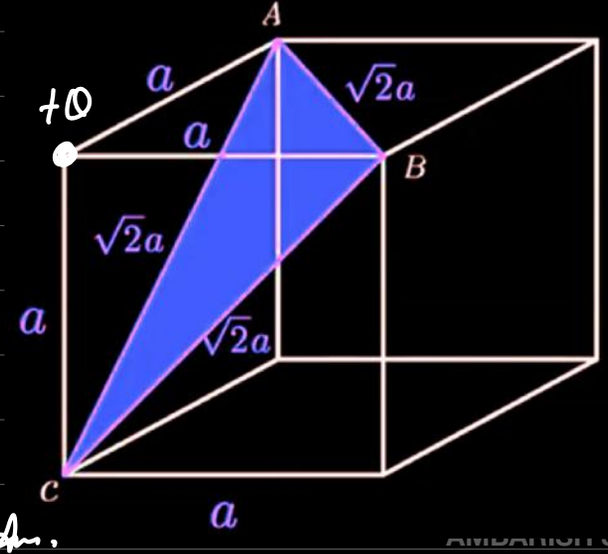
$\therefore$ Option (d)

EXAMPLE: Base of a tetrahedron is an equilateral triangle, while the other three faces are isosceles right triangles as shown in the figure. If the base has a surface charge density σ, and apex carries a charge $+Q$. Calculate the force of interaction between the base and $+Q$.



Solution: Imagine the charge $+Q$ to be a small sphere centred at the apex then 1/8 of that sphere would be enclosed in the tetrahedron. Thus flux through tetrahedron is $Q/8\epsilon_0$. All of this flux must be through equilateral face because field lines are parallel to isosceles faces. $\therefore$

$$F = \sigma\phi = \frac{Q}{8\epsilon_0} \times \sigma \Rightarrow F = F_N = \frac{Q\sigma}{8\epsilon_0}$$ Ans.



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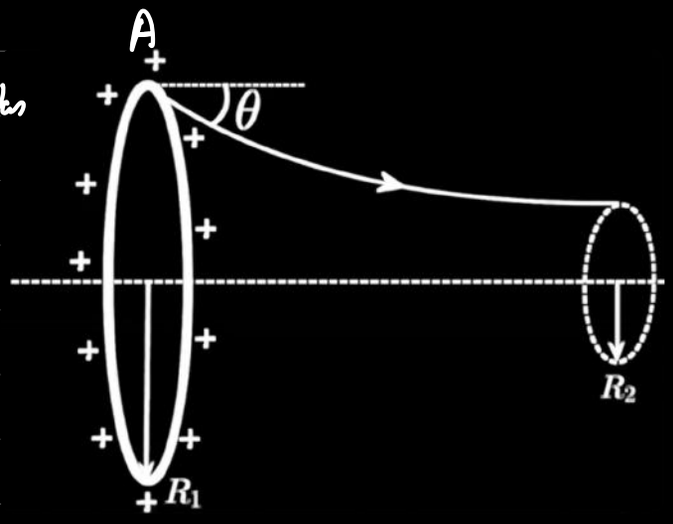
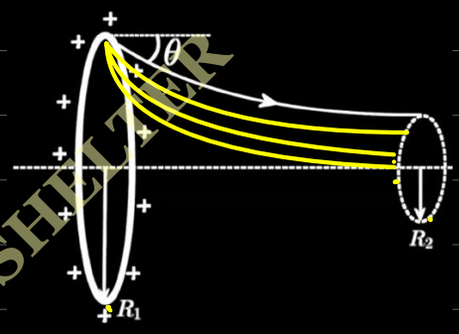
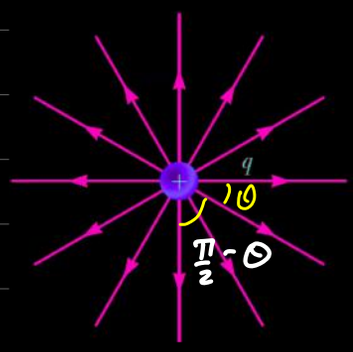
EXAMPLE: A disc has charge density σ while a ring has charge density λ_0. The ring and disc are coaxial. A particular field line grazing past the periphery of the disc is shown. Find the force of interaction between the ring and the disc.



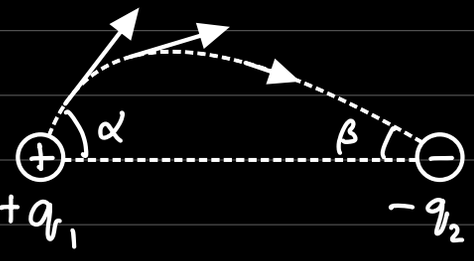
Solution: Just like with point charges, with line segment charges field is infinite at zero distance from them. (This is not the case with area charges). Thus at very near points, field lines are not affected by remote elements. Now imagine a small line element near the top of the ring, one can easily see that field lines emanating within an angle of $90°-\theta$ pass through the disc. If its charge were dq then because of this flux entering the disc is $\frac{(\pi/2-\theta)}{2\pi} dq/\epsilon_0$. Thus total flux through the disc will be $\left(\frac{\pi/2-\theta}{2\pi}\right)\frac{Q}{\epsilon_0}$ and accordingly the force is given by $F_N = \Phi \times \sigma = \left(\frac{1}{4} - \frac{\theta}{2\pi}\right)\frac{Q\sigma}{\epsilon_0} = \frac{\lambda_0 \times 2\pi R_1 \sigma}{\epsilon_0}\left(\frac{1}{4} - \frac{\theta}{2\pi}\right)$





$$\Rightarrow F = \frac{\lambda_0 \sigma}{\epsilon_0} R_1 \left[\frac{\pi}{2} - \theta\right] \text{ Ans.}$$

EXAMPLE: KROTOV 3.2 : An electric field line emerges from a point charge q_1 at an angle α to the straight line connecting it to charge $-q_2$. At what angle β will the field line enter the charge $-q_2$.



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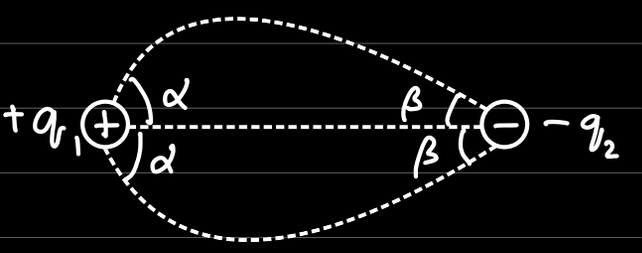
Solution: Key Idea here is that at points very near to point charge the contribution of field due to remote charges is negligible as compared to the point charge itself.



Also since field lines never intersect each other therefore field lines emitted by q_1 in solid angle corresponding to semi vertex angle α must be equal to the number field lines entering $-q_2$ in a semi vertex angle β.

Thus: $$\frac{q_1}{\epsilon_0}\,\frac{2\pi(1-\cos\alpha)}{4\pi} = \frac{q_2}{\epsilon_0} \times \frac{2\pi(1-\cos\beta)}{4\pi} \quad (1)$$

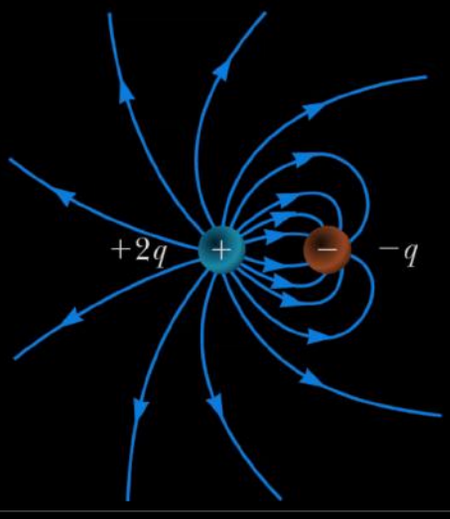
$$\Rightarrow \cos\beta = 1 - \frac{q_1}{q_2}(1-\cos\alpha) \text{ Ans.}$$

THINK YOURSELF: If $q_1 > q_2$ what would be the maximum value of α so that a field line emitted at α does enter $-q_2$?

Solution: See adjacent figure. $\beta_{max} = 180^\circ$ and α_{max} must correspond to β_{max}. Thus, putting $\beta_{max} = 180^\circ$ in equation (1), we get

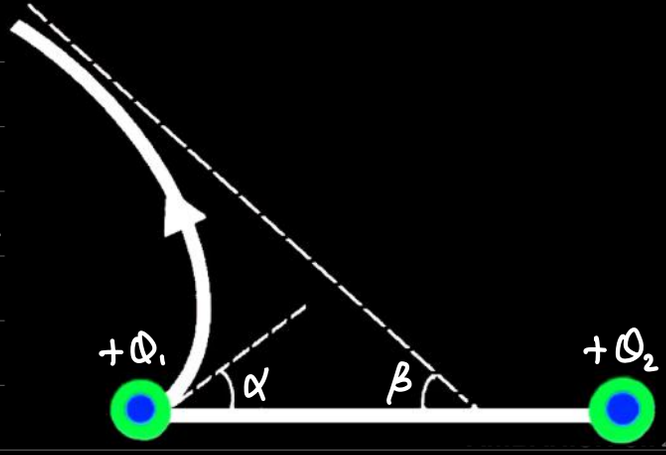
$$q_1(1-\cos\alpha_{max}) = 2q_2$$

$$\Rightarrow \cos\alpha_{max} = 1 - \frac{2q_2}{q_1} \text{ Ans.}$$



THINK, For a dipole, $q_2 = q_1 \Rightarrow \alpha_{max} = 180^\circ$
$\Rightarrow$ Each field line emitted by q_1 enters q_2.

EXAMPLE: Consider two positive charges Q_1 and Q_2 as shown in the figure. A tangent is drawn at ∞ to a certain field line emanating from Q_1 at an angle α as shown. The tangent makes an angle β with line joining the charges as shown. Find the relation between Q_1, Q_2, α and β.



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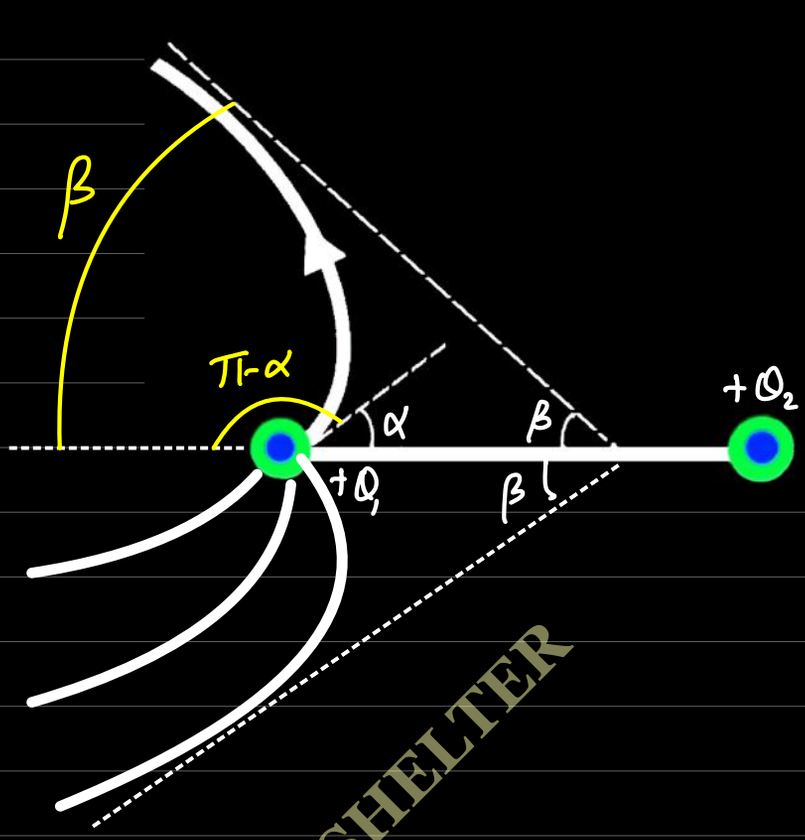
Solution: From ∞ the system of two point charges appears to be just a single point charge $Q_1 + Q_2$. Also every field line emitted by Q_1 in a cap of semivertex angle $\pi - \alpha$ enters a cap of semivertex angle β at ∞ (∵ field lines can't intersect each other).



Thus:

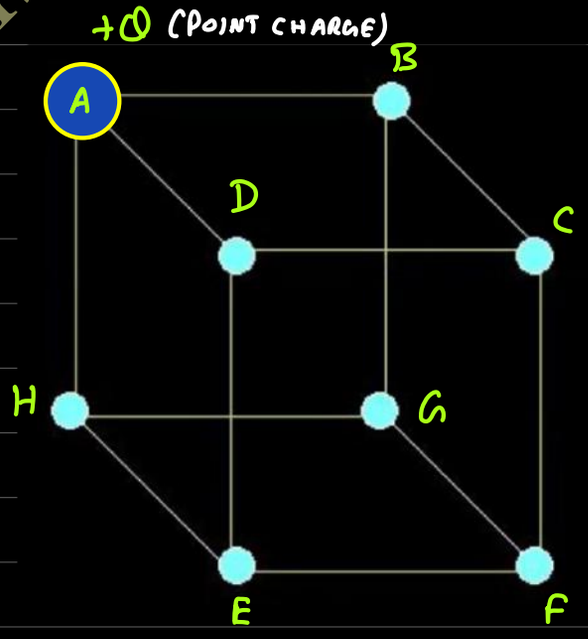
$$\frac{Q_1}{2\epsilon_0}(1-\cos(\pi-\alpha)) = \frac{Q_1+Q_2}{2\epsilon_0}(1-\cos\beta)$$

$$\Rightarrow Q_1(1+\cos\alpha) = (Q_1+Q_2)(1-\cos\beta) \Rightarrow Q_1(\cos\alpha+\cos\beta) = Q_2(1-\cos\beta)$$ Ans.

EXAMPLE: At one of the vertices of a Gaussian Cube, a charge $+Q$ is placed. Find the flux through three faces nearest to the charge if now the charge is shifted:

(a) Slightly outside the cube along the face diagonal CA

(b) Slightly inwards along body diagonal AF.



Solution: When the point charge is exactly at the vertex A, exactly 1/8 of it lies within the cube, thus flux through the entire cube is $Q/8\epsilon_0$. Out of this flux through three nearby surfaces must be zero (∵ field lines ⊥ to faces), thus entire flux is through the farther three faces (Each one has $Q/24\epsilon_0$ by symmetry).

Now when Q is shifted very slightly, the flux through farther faces remains almost unchanged. Now:

(a) When Q is pushed out of the cube, net flux through cube becomes zero. ∴ Flux through three nearby faces must add up

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to $-Q/8\epsilon_0$ ($\because$ Through far face is $+Q/8\epsilon_0$). Out of this flux through ABCD must be still zero $\because$ charge still lies in this plane, and by symmetry the flux through other two nearby faces must be same. Thus:

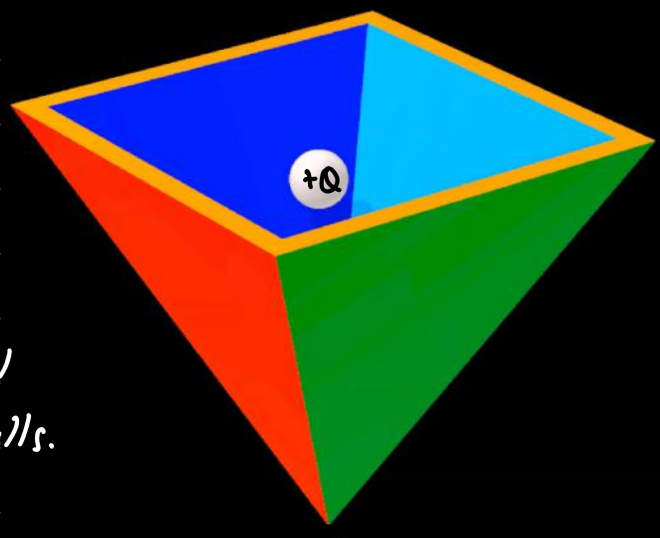
$$\Phi_{ABCD} = 0, \quad \Phi_{ADEH} = -Q/16\epsilon_0 \text{ and } \Phi_{ABGH} = -Q/16\epsilon_0.$$

(b) When charge is moved inward along body diagonal, net flux through cube becomes Q/ϵ_0. Now since flux through three far away faces is still $Q/8\epsilon_0$ $\therefore$ flux through three nearby faces must be $7Q/8\epsilon_0$ By symmetry this is equally divided in to three nearby faces. Thus:

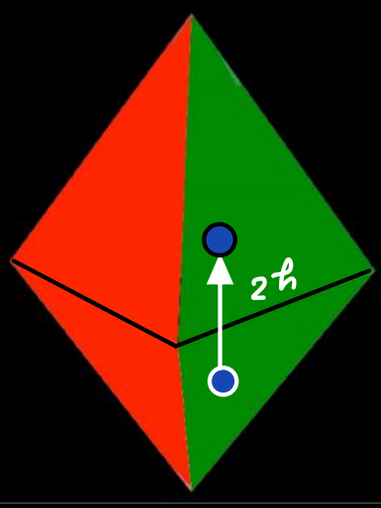
$$\Phi_{ABCD} = \Phi_{ADEH} = \Phi_{ABGH} = 7Q/24\epsilon_0 \text{ Ans.}$$

THINK YOURSELF: What about the cases when you shift Q along edge (outside or inside), What about shifting outside along body diagonal.

EXAMPLE: A point charge Q at a depth of h from square base of a pyramid shell produces a flux of $\frac{2Q}{3\epsilon_0}$ from the four walls of the shell as shown in the figure. If now the charge is raised through a height 2h, what will be the new flux through the four walls.



Solution: Imagine Completing the closed Now the flux through complete Gaussian surface must be Q/ϵ_0. This means flux through upper pyramid must be $Q/3\epsilon_0$. Now if the charge is raised through height 2h its location wrt upper pyramid is similar to how it was with lower pyramid before shifting, thus now fluxes are also swapped.



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Thus now flux through upper pyramid is $2Q/3\epsilon_0$ ∴ flux through lower pyramid is $Q/3\epsilon_0$ obvs.

→ Note that instead of completed the Gaussian Surface with congruent pyramid, we could have completed the surface with some other shape as well and it would make no difference to our argument.

Field Due To A Spherical Shell:

Consider a spherical shell of radius R having a surface charge density σ. Using Gauss law we shall prove the following:

(a) Field inside the shell ($r < R$) is

$$\boxed{E(r) = 0}$$

(b) Field outside the shell ($r > R$) is given by:

$$\boxed{E(r) = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2} = \frac{\sigma}{\epsilon_0}}$$

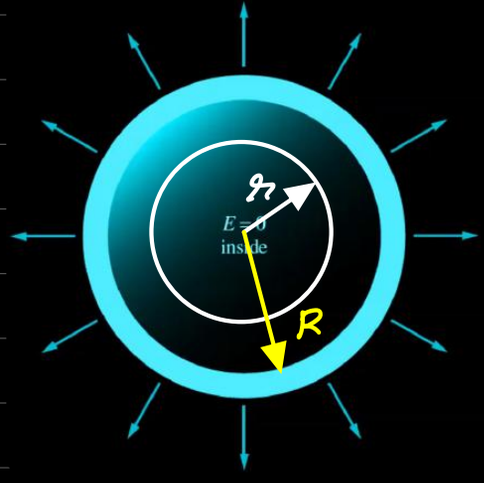
→ These facts are also known as shell theorems:

Shell Theorem I: For an inside point a uniformly charged shell produces no field.

Shell Theorem II: For an outside point a shell behaves just like a point charge placed at its own centre.

Proof Of Shell Theorem I

Method I: Using Gauss Law.



Consider a Gaussian spherical surface of radius r ($< R$) as shown. Applying Gauss law:

$$E \times 4\pi r^2 = Q/\epsilon_0 \quad (\because \text{No charge enclosed})$$

$$\Rightarrow E = 0 \quad \text{QED.}$$

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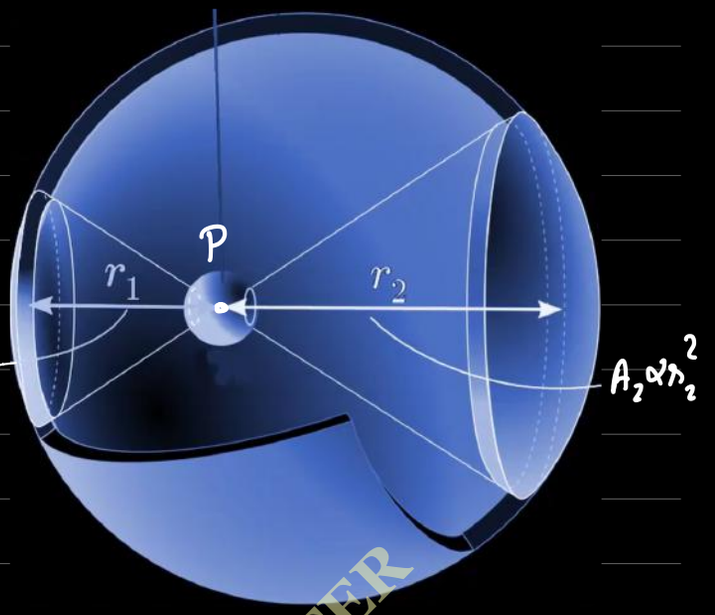
ALTERNATE PROOF OF SHELL THEOREM I :

Consider a point P inside the shell, The net field at Point P is due to resultant of fields due to patches A_1 and A_2. Now point P is at the apex of similar cones.



Net field at P:

$$E = \frac{k\sigma A_1}{r_1^2} - \frac{k\sigma A_2}{r_2^2}$$

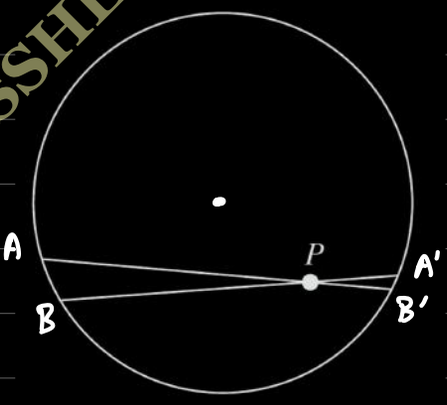
But due to similarity $A_1/r_1^2 = A_2/r_2^2$

$\Rightarrow E = 0.$

Even if we were not to form right circular cones, we can still see that the two patches will produce cancelling fields. We can prove that cones defined by APB and A'PB' are similar. By chord intersection theorem:

$$AP \times PB' = A'P \times PB$$

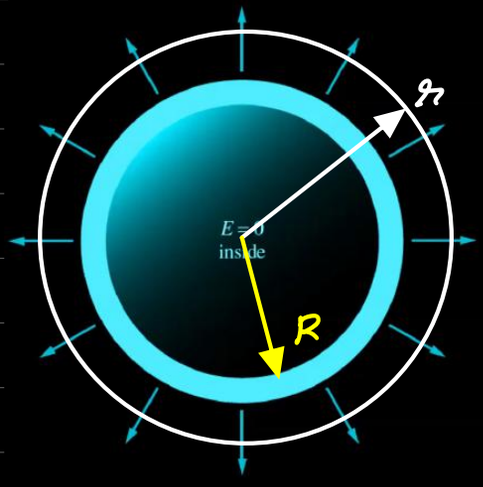
$$\Rightarrow \frac{AP}{A'P} = \frac{PB}{PB'}, \text{ Also } \angle APB = \angle A'PB'$$



Thus the cones are similar and areas are therefore proportional to squares of linear distances.

PROOF OF SHELL THEOREM II :

Consider a Gaussian sphere outside the shell. Now applying Gauss theorem:



$$E \times 4\pi r^2 = Q/\epsilon_0$$

$$\Rightarrow E = \frac{1}{4\pi\epsilon_0}\frac{Q}{r^2} = \frac{\sigma}{\epsilon_0}$$

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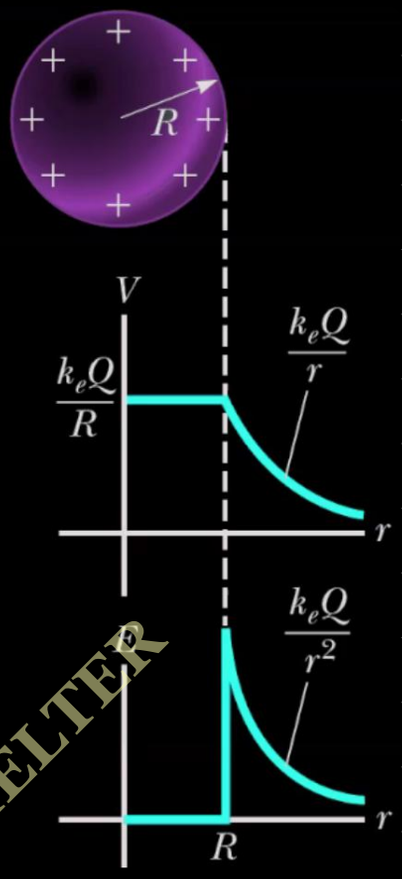
Potential Due To A Spherical Shell:



Since for an outside point a shell behaves just like a point charge, ∴ its potential at outside point is also just like that for a point charge. Thus for $r > R$,

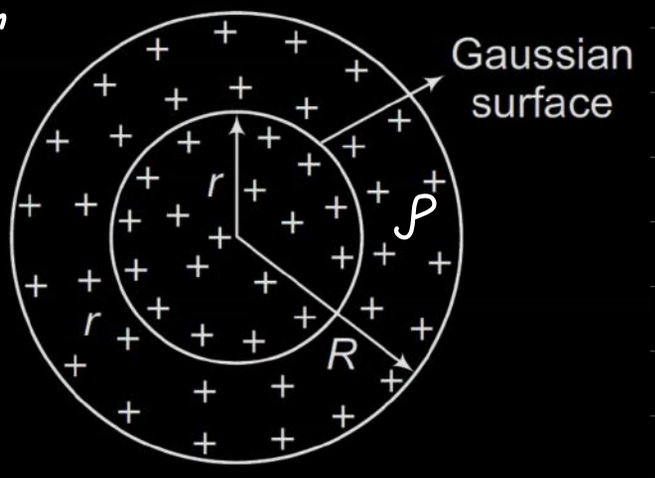
$$\boxed{V = kQ/r}$$

Now since field inside the shell is zero, ∴ there is no work done in coming from surface to an inside point. ∴ Potential inside is same as the potential at the surface of the shell. i.e for inside point. ($r < R$):

$$\boxed{V = kQ/R}$$

Adjacent figure shows variation of field and potential due to a shell.

Field Due To A Uniformly Charged Solid Sphere:



Consider a solid sphere having a uniform charge density ρ and a total charge Q and a radius R. A solid sphere can always be thought of as made of several thin concentric shells, thus by shell theorems, field and potential at outside points are just like that of a point charge. i.e for $r \geq R$:

$$\boxed{E = \frac{kQ}{r^2}} \qquad \boxed{V = \frac{kQ}{r}}$$

For an inside point, we can find field using Gauss Theorem: Consider the Gaussian surface shown in the figure ($r < R$), thus:

$$E \times 4\pi r^2 = \rho \times \frac{4}{3}\pi r^3/\epsilon_0 \implies E = \rho r/3\epsilon_0$$

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Vectorially, for $r < R$, $\boxed{\vec{E} = \frac{\rho \vec{r}}{3\epsilon_0}}$

Also $\because \rho = Q/\frac{4}{3}\pi R^3 \Rightarrow \boxed{\vec{E} = \frac{kQ\vec{r}}{R^3}}$

Now we know that $V(R) = kQ/R$ and also, $E = -dV/dr$, we can find V.

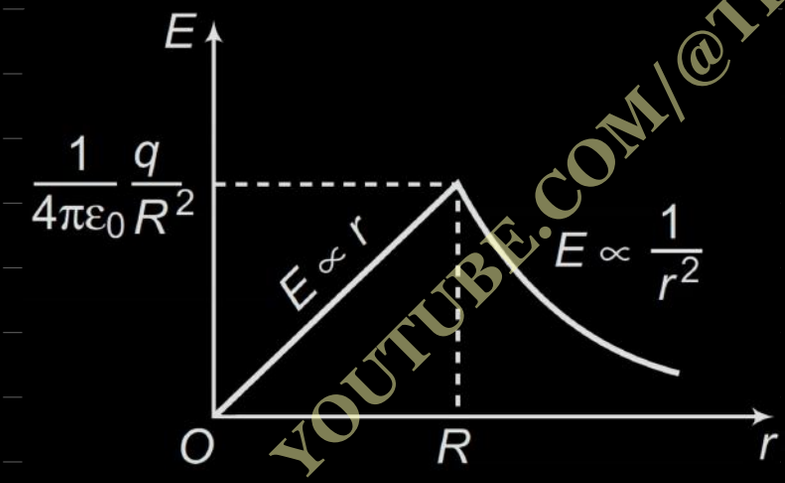
$$\frac{dV}{dr} = -\frac{kQr}{R^3} \Rightarrow V = -\frac{kQr^2}{2R^3} + C$$

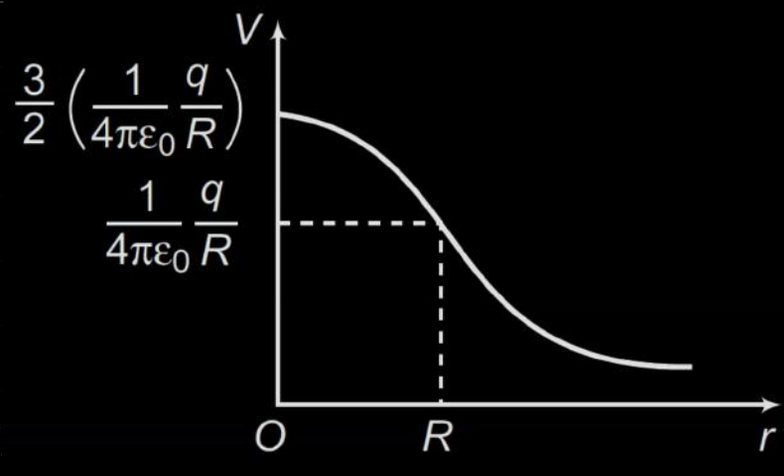
Also @ $r = R$, $V = \frac{kQ}{R} \Rightarrow \frac{kQ}{R} = -\frac{kQR^2}{2R^3} + C$

$$\Rightarrow C = \frac{3}{2}\frac{kQ}{R} \Rightarrow V = -\frac{kQr^2}{2R^3} + \frac{3}{2}\frac{kQ}{R}$$

$$\Rightarrow \boxed{V = \frac{kQ}{R}\left[1.5 - 0.5r^2/R^2\right]}$$

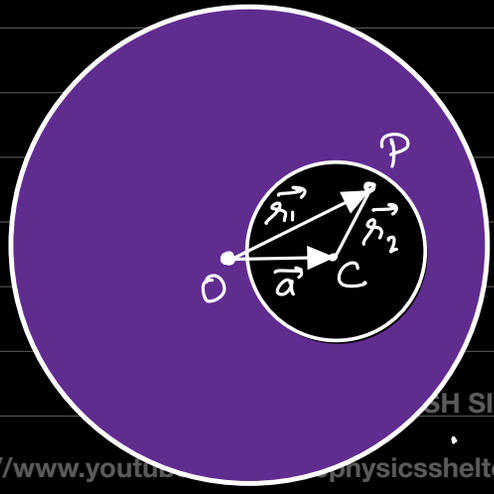
Plot Of Field & Potential For A Solid Sphere:





Field In A Spherical Cavity In A Uniformly Charged Sphere:

Let P be an arbitrary point in a cavity inside a spherical distribution of charge density ρ. C is centre of cavity and O is centre of sphere. Thus $\vec{E}$:



$$\vec{E} = \frac{\rho\vec{r}_1}{3\epsilon_0} - \frac{\rho\vec{r}_2}{3\epsilon_0} = \frac{\rho(\vec{r}_1 - \vec{r}_2)}{3\epsilon_0}$$

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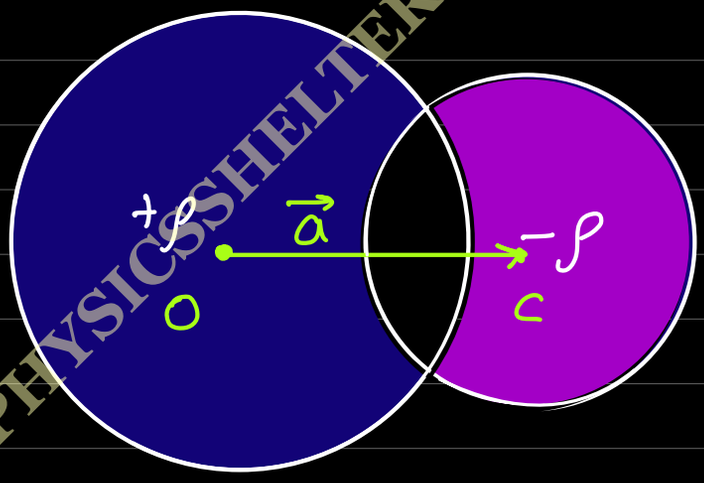
$\Rightarrow \boxed{\vec{E} = \frac{\rho \vec{a}}{3\epsilon_0}}$ Here since removed portion was also spherical, we used the formula for a solid sphere to compensate for the missing spherical charge.

→ Note that field in this cavity is uniform and parallel to line joining the two centres. Its magnitude depends on separation between two centres.

→ Alternatively we can also visualise the cavity as superposition of equal and opposite charge densities (ρ and $-\rho$).

THINK YOURSELF: A positively charged sphere is superposed on a negatively charged sphere as shown. What will be the field in the neutral area.

Ans. $\rho\vec{a}/3\epsilon_0$



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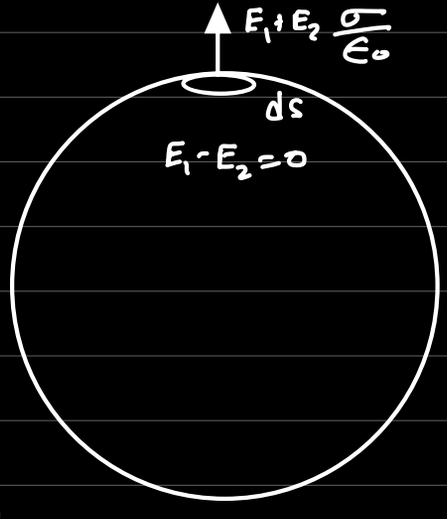
ELECTROSTATIC PRESSURE ON THIN SPHERICAL SHELL (UNIFORMLY CHARGED)

Consider a small patch on surface of a thin uniformly charged spherical shell. Let its field contribution in its vicinity be E_1 and due to rest of the shell be E_2. Then at at point just outside, since net field is σ/ϵ_0 $\therefore$ $E_1 + E_2 = \sigma/\epsilon_0$ ① Similarly inside field is zero $\therefore$ $E_1 - E_2 = 0$ ② From ① & ②, $E_1 = E_2 = \sigma/2\epsilon_0$

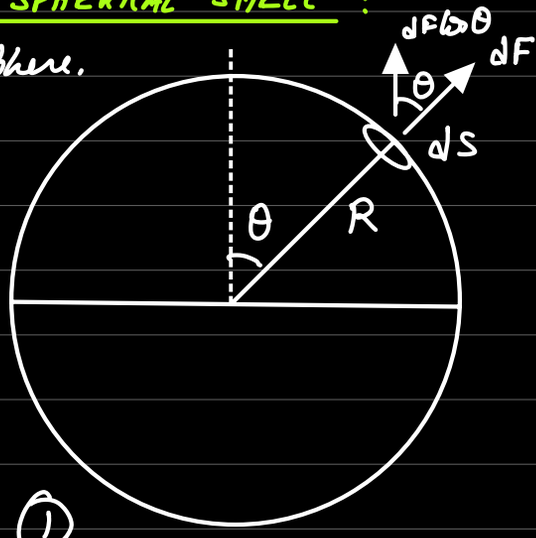


Now Force on this patch due to rest of the shell,

$$dF = \sigma\, dS \times E_2 = \frac{\sigma^2}{2\epsilon_0} dS$$

$\Rightarrow$ Pressure: $\frac{dF}{dA} = \frac{\sigma^2}{2\epsilon_0}$ $\boxed{P = \sigma^2/2\epsilon_0}$

FORCE OF REPULSION BETWEEN TWO HALVES OF SPHERICAL SHELL :



Consider a small patch dS of the upper hemisphere.

$dF = \frac{\sigma^2}{2\epsilon_0} dS$, its vertical component :

$$dF_v = dF\cos\theta = \frac{\sigma^2}{2\epsilon_0} dS\cos\theta.$$

Now realise that $dS\cos\theta$ is shadow area of dS in base of hemisphere. Thus,

$$dF_v = \frac{\sigma^2}{2\epsilon_0} . dS_{shadow} \Rightarrow F_v = \frac{\sigma^2}{2\epsilon_0} . \pi R^2 \quad (1)$$

$$\Rightarrow F = \frac{1}{2\epsilon_0} \cdot \left(\frac{Q}{4\pi R^2}\right)^2 . \pi R^2 = \frac{1}{32\pi\epsilon_0} \frac{Q^2}{R^2}$$

Note that dF includes internal forces, however internal forces cancel when summation is taken over the entire upper hemispherical shell.

Alternate: $d\vec{F} = \frac{\sigma^2}{2\epsilon_0} . dS\,\hat{R} = \frac{\sigma}{2\epsilon_0 R} \vec{R}\, dq$ $(\because \sigma dS = dq)$

$$\Rightarrow \vec{F} = \frac{\sigma}{2\epsilon_0 R}\left(\int \vec{R}\, dq\right) \times \frac{Q/2}{(Q/2)} = \frac{\sigma Q}{4\epsilon_0} \times \vec{r}_{CQ} \quad (\because \text{Centre of Charge})$$

$$= \frac{\sigma Q}{4\epsilon_0 R} \cdot \frac{R}{2}\,\hat{n} \quad (\hat{n} \text{ is vertical upwards}).$$

$$= \frac{Q^2}{32\pi\epsilon_0 R^2}\hat{n} \quad (\text{This is same result as obtained earlier}).$$

→ If instead of same charge density σ on upper and lower hemispheres if it were σ_1 and σ_2 on upper and lower hemispheres, then σ^2 will have to be replaced by $\sigma_1\sigma_2$ in equation (1), because every pairwise interaction will be scaled by a factor of $\sigma_1\sigma_2/\sigma^2$. ($\because$ same scale factor applies to overall force as well).

→ If there were a small hole in the hemispherical shell, the field in the hole would be $\sigma/2\epsilon_0$ ($\because$ only field due to rest of the hemispherical shell would be present there).

EXAMPLE: Find the work done by electric forces in changing the radius of a uniformly charged spherical shell from R_1 to R_2,

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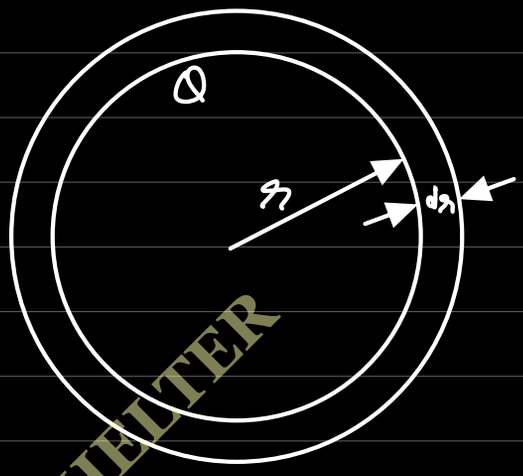
hence find the work done by electric forces in forming the spherical shell bringing all its charges from ∞. What is potential energy of this configuration if at ∞ potential energy is assumed to be zero.

Solution: $P = \frac{\sigma^2}{2\epsilon_0} = \frac{Q^2}{32\pi^2\epsilon_0 r^4}$, $dV = 4\pi r^2 dr$



$\Rightarrow PdV = \frac{Q^2}{8\pi\epsilon_0 r^2} dr$

$\Rightarrow \int dW = \frac{Q^2}{8\pi\epsilon_0} \int_{R_1}^{R_2} \frac{1}{r^2} dr$

$\Rightarrow \Delta W = \frac{Q^2}{8\pi\epsilon_0} \left[\frac{1}{R_1} - \frac{1}{R_2}\right]$ Ans.

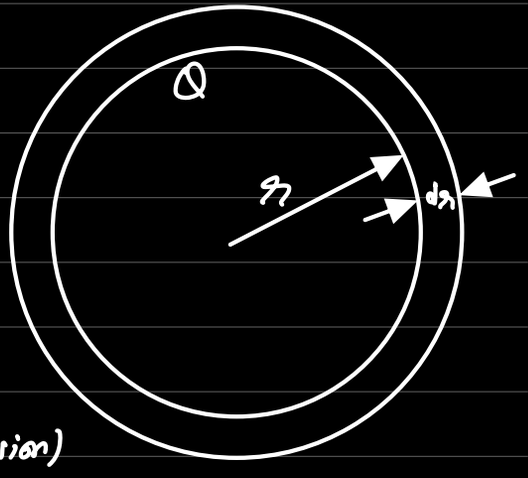
Now to make the sphere from ∞ to R, $R_1 = \infty$, $R_2 = R$ $\Rightarrow$

$\Delta W_{Electric} = \frac{-Q^2}{8\pi\epsilon_0 R}$ $\Rightarrow \Delta U = U = \frac{Q^2}{8\pi\epsilon_0 R}$

$\Rightarrow$ $\boxed{U = \frac{kQ^2}{2R}}$ ← SELF ENERGY OF A THIN SPHERICAL SHELL.

→ When all charges were at ∞, there was no electric field in the universe even though infinitesimal charged particles were there, when they were brought together, universe got some field. Thus this potential energy is stored in electric field. (As if some invisible springs got pulled from ∞ as charges were brought from ∞).

ENERGY DENSITY IN SPACE : When a shell expands from r to r+dr, the field in rest of the space remains as it is, however it vanishes in an additional volume $4\pi r^2 dr$. ∴ We must deem that Energy change associated with work of electrostatic pressure must be associated with this volume. Let energy stored in this volume be dE (before expansion) and Energy in rest of space be E_0



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Now $P\,dV = \frac{\sigma^2}{2\epsilon_0}\cdot dV = -dU = -[E_0 - (E_0 + dE)] = dE$

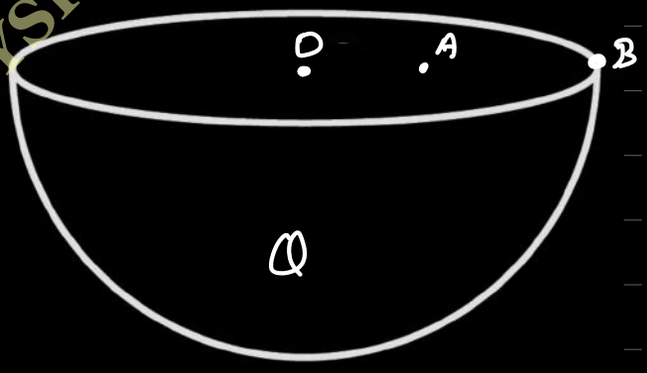
$\Rightarrow \frac{dE}{dV} = \frac{\sigma^2}{2\epsilon_0}$, Also $E = \frac{\sigma}{\epsilon_0} \Rightarrow \frac{dE}{dV} = \frac{1}{2}\epsilon_0 E^2$

Thus energy density associated with volume is $\boxed{u = \frac{1}{2}\epsilon_0 E^2}$,

Note that we did not assume any particular radius for proving this result for proving this expression, thus everywhere in space, energy density is $\frac{1}{2}\epsilon_0 E^2$.

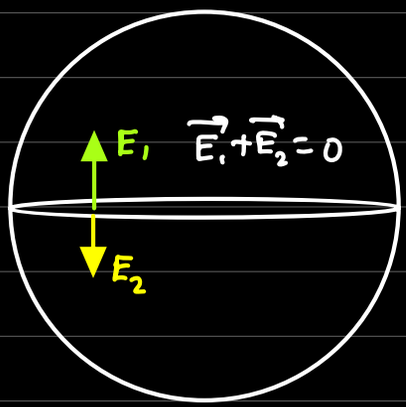
→ Even though we have proved this result for a specific geometry, the result of energy density in space is in general true. i.e if E is the local electric field, energy density in space is $u = \frac{1}{2}\epsilon_0 E^2$.

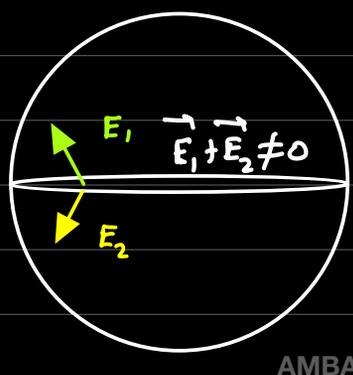
EXAMPLE: A hemispherical shell has uniform charge Q spread on it. How do the potentials at O, A and B compare.



Solution: If we place a similar hemisphere to complete the spherical shell, then field inside must be zero by shell theorem, thus field due to lower shell must exactly cancel the field due to upper hemispherical shell at every point in the equatorial plane. This means that in the equatorial plane, everywhere, field must be ⊥ to equatorial plane. Otherwise fields can't cancel by symmetry. (See figure below). Thus the equatorial plane must be an equipotential surface.

Thus $V_B = V_A = V_O = \frac{kQ}{R}$ Ans.





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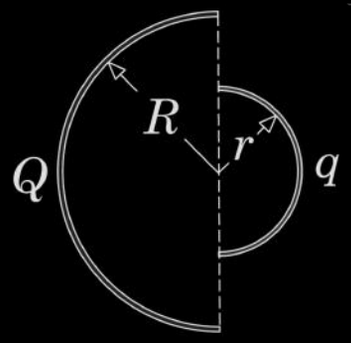
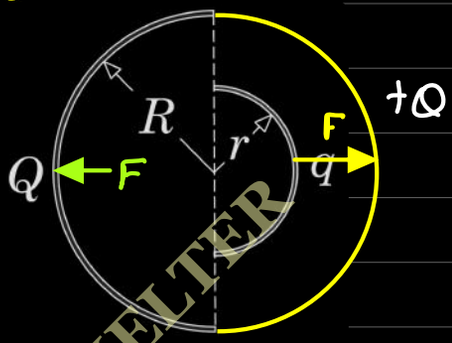
EXAMPLE: PATHFINDER CHECK-6 .

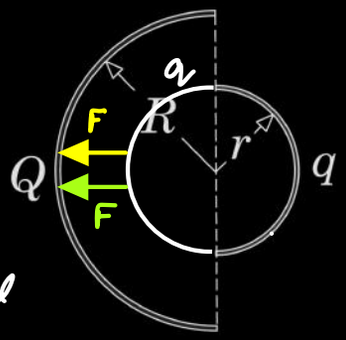
Two hemispherical thin shells made of insulating materials are concentrically arranged in a free space as shown. The radii of the larger and the smaller hemispheres are R and r and they carry net positive charges Q and q respectively. The charges are uniformly distributed over the surfaces of the shells. Find the force of electrostatic interaction between the shells.



Solution: Consider completing the outer shell, then force must become zero by shell theorem. Thus both the outer shells exert a force of equal magnitude on inner hemispherical shell .: inner hemispherical shell must also exert equal magnitude forces on left and right outer hemispheres. Let this force be F.

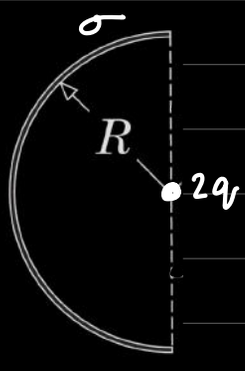


Now Consider completing the inner shell, now the interaction force must be 2F from the argument of the previous figure. Thus our required force must be half of the force exerted by complete spherical shell of radius r. Now this spherical shell behaves like a a point charge at its centre for purpose of finding force on external objects (or finding force of external objects on it.

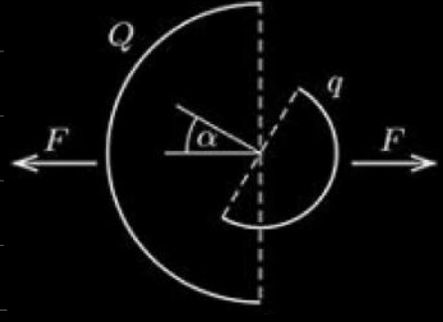


Now recall field due to hemispherical shell at its centre of curvature is $\sigma/4\epsilon_0$. Thus if F is our required force then $2F = \frac{\sigma}{4\epsilon_0} \times 2q$

where $\sigma = Q/2\pi R^2 \Rightarrow F = \sigma q/4\epsilon_0 = \frac{Qq}{8\pi R^2 \epsilon_0}$ Ans.



→ By a similar argument it can be shown that our answer will remain same even if our smaller shell were turned through an angle α about its centre of curvature.



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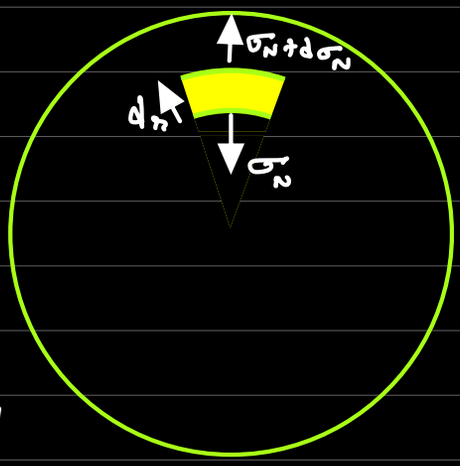
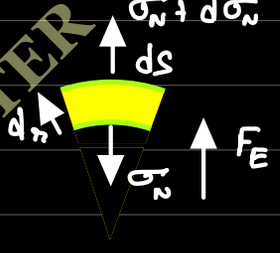
Electrostatic Pressure (& Material Stress) in a Solid Charged Sphere:

Consider a solid sphere having a uniform charge density ρ. Due to electrostatic forces of its own repulsion, the sphere has a tendency to explode, however it is prevented from doing so due to tensile stresses between its layers. Consider a layer element of thickness dr having an area ds. Let its distance from centre be r. Now the local field is given as $\vec{E} = \frac{\rho \vec{r}}{3\epsilon_0}$. Thus balancing forces on the layer:





$$\frac{\rho r}{3\epsilon_0} \times (ds \times dr)\rho + (\sigma_N + d\sigma_N)\, ds = \sigma_N\, ds$$

$$\Rightarrow \quad \frac{\rho^2 r}{3\epsilon_0} dr = -d\sigma_N \quad (1)$$

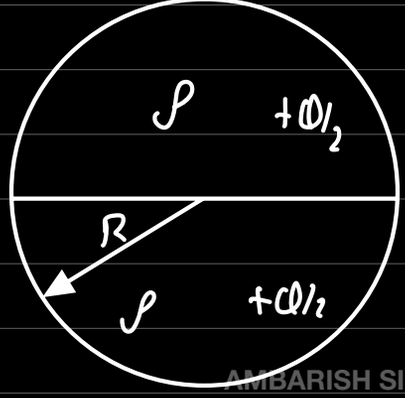
Also @ $r = R$, $\sigma_N = 0$ since there is no material beyond $r = R$ to create any pulling force. Now integrating (1) with proper limits.

$$\frac{\rho^2}{3\epsilon_0} \int_R^r r\, dr = -\int_0^{\sigma_N} d\sigma_N$$

$$\Rightarrow \quad \boxed{\sigma_N = \frac{\rho^2}{6\epsilon_0}(R^2 - r^2)}$$

→ This is the tensile stress developed in response to electrostatic repulsion and therefore also known as electrostatic pressure (Pressure has same units as stress). In equilibrium, electrostatic repulsion exactly balances the difference of tensile forces across the element.

Example: Find the force of electrostatic repulsion between two halves of a solid hemisphere carrying uniform charge density ρ and having Radius R (Total charge is Q).



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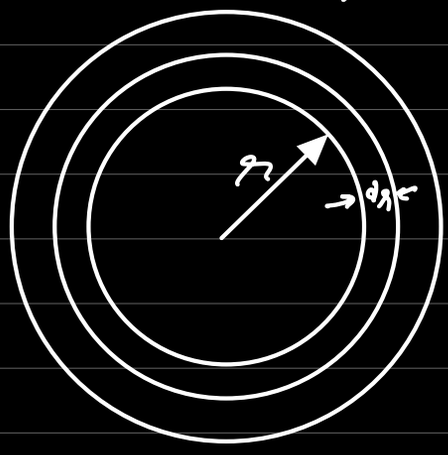
Solution: Consider FBD of upper hemisphere. Net electrostatic repulsion acting on its volume must be exactly equal to tensile pulling force on its base for equilibrium. Now consider an annular region in the equatorial plane.



$$dF = \sigma_N \times 2\pi r\,dr$$

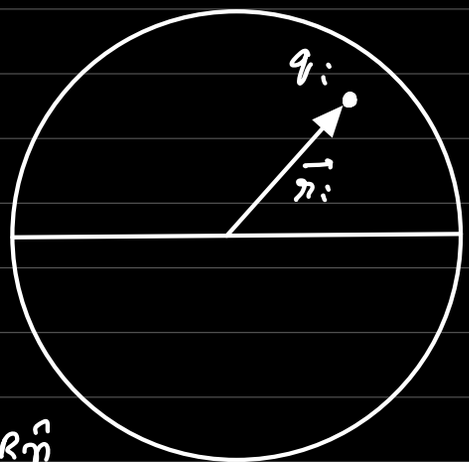
$$= \frac{\rho^2}{6\epsilon_0} \cdot (R^2 - r^2) \times 2\pi r\,dr$$

$$\Rightarrow F = \frac{\pi\rho^2}{3\epsilon_0} \int_0^R (R^2 r - r^3)\,dr$$

$$= \frac{\pi\rho^2 R^4}{12\epsilon_0} = \frac{3Q^2}{64\pi R^2\epsilon_0} \quad \text{Ans.}$$

→ Note that if the upper and lower charge densities were different, we could arrive at the result by replacing ρ^2 with $\rho_1\rho_2$ or Q^2 with $4Q_1Q_2$ (Since Q_1, Q_2 corresponds to $\frac{Q}{2} \times \frac{Q}{2}$ ∵ charge on each hemisphere was $Q/2, Q/2$).

ALTER: Consider a particle of charge q_i at position vector $\vec{r}_i$ in the upper hemisphere. The electrostatic force on this is given as



$$\vec{F}_i = \frac{\rho\vec{r}_i}{3\epsilon_0} \cdot q_i$$

$$\Rightarrow \vec{F} = \Sigma\vec{F}_i = \frac{\rho}{3\epsilon_0}\left(\Sigma q_i \cdot \vec{r}_i\right) \times \left(\frac{Q/2}{Q/2}\right)$$

$$= \frac{\rho Q}{6\epsilon_0}\vec{r}_{CO} = \frac{\rho Q}{6\epsilon_0} \cdot \frac{3R}{8}\hat{n} = \frac{\rho Q R\hat{n}}{16\epsilon_0}$$

Now using $Q = \rho \times \frac{4}{3}\pi R^3$, $\quad F = \frac{3Q^2}{64\pi R^2\epsilon_0} = \frac{\pi\rho^2 R^4}{12\epsilon_0}$ Ans.

→ Same technique could be used for calculating force on any portion of the sphere (you will need to find centre of charge and charge of the portion).

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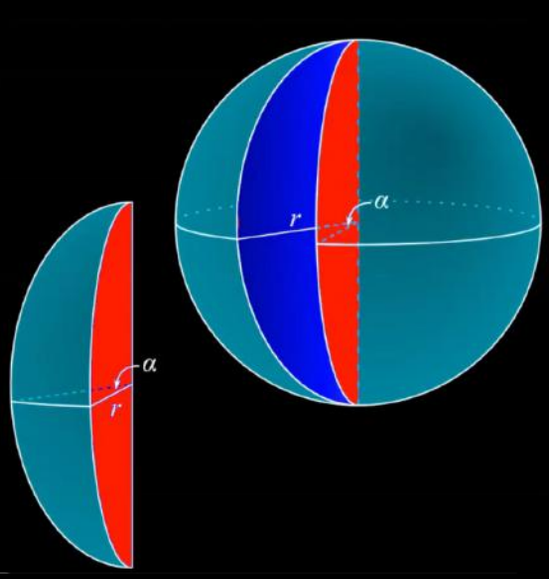
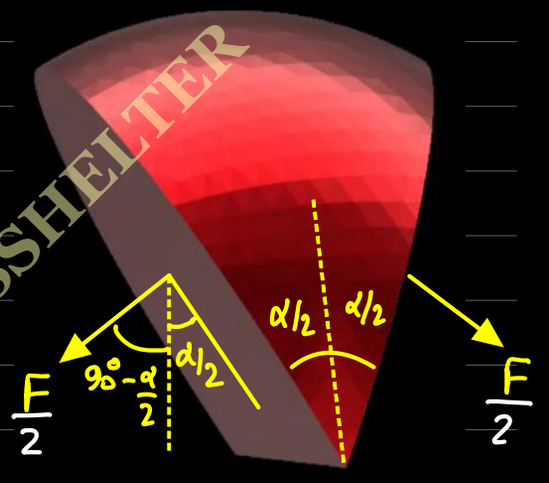
EXAMPLE: If F is the force between two solid hemispheres, what would be the electrostatic force on a wedge of angle α.



Solution: A wedge has two semicircular faces. The force between two hemispheres is integral of pressure over entire circle of radius R. Thus the integral on a semicircular face of radius R will be F/2. Now consider the FBD of the wedge, The vertical component of tensile force on one of the faces is $\frac{F}{2}\sin(\alpha/2)$.

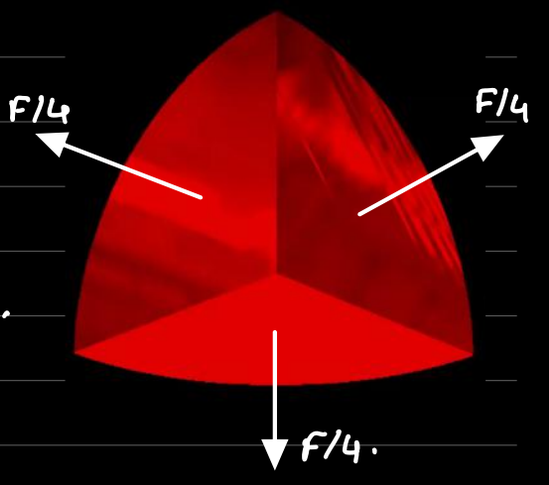


Thus net force on the wedge :

$$\vec{F}_w = \frac{F}{2}\sin(\alpha/2) \times 2 = F\sin(\alpha/2)$$

→ Think Yourself: What would be the force on octant of a sphere? Ans: $\frac{F}{4}\sqrt{3}$

Solution: Octant of a sphere will have three mutually perpendicular quarter circular faces thus net force $\frac{F}{4}(\hat{i}+\hat{j}+\hat{k}) \Rightarrow F_{net} = \frac{F}{4}\sqrt{3}$ Ans.



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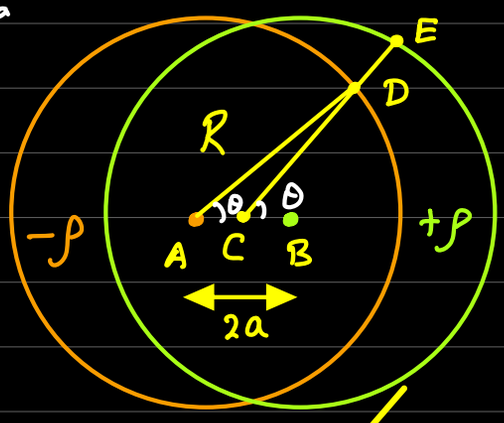
EXAMPLE: IRODOV 3.17

Suppose the surface charge density over a sphere of radius R depends on a polar angle θ as $\sigma = \sigma_0 \cos\theta$, where σ_0 is a positive constant. Show that such a charge distribution can be represented as a result of a small relative shift of two uniformly charged *balls* of radius R whose charges are equal in magnitude and opposite in sign. Resorting to this representation, find the electric field strength vector inside the given sphere.

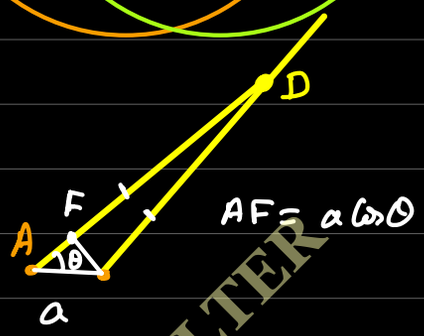
Solution: Consider a positively charged and a negatively charged sphere having charge densities $+\rho$ and $-\rho$ respectively. Each has a radius R and centre to centre separation is $2a$. ($a \to 0$, figure is highly exaggerated). $\because a \to 0$, this charge distribution would effectively look like a surface charge distribution. Overlapping portion has obviously zero charge. We shall show that thickness of the charged portion is proportional to $\cos\theta$.



Now $CD = AD - a\cos\theta = R - a\cos\theta$

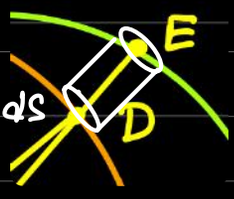
Similarly $CE = BE + a\cos\theta = R + a\cos\theta$,

Thus thickness of the charged layer $DE = CE - CD = 2a\cos\theta$.



Now imagine a cylinder of cross section dS with axis along DE. The charge spread in this thickness is given as, $dq = (dS \times 2a\cos\theta) \times \rho$

$\Rightarrow \sigma = \frac{dq}{dS} = 2\rho a\cos\theta$, notice that this



is exactly the same form as $\sigma = \sigma_0\cos\theta$ where $\sigma_0 = 2\rho a$.

Now field in the overlapping region is given as

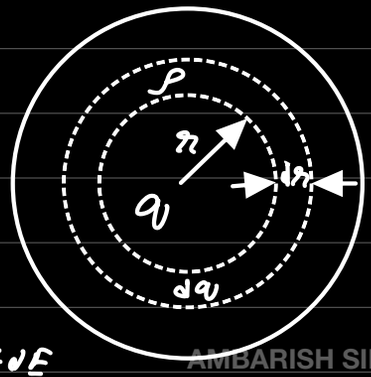
$$E = \rho\frac{2a}{3\epsilon_0} = \frac{\sigma_0}{3\epsilon_0}$$ ($\because$ Recall field in cavity / overlapping +/- spheres).

→ For inside points, there is uniform field

→ For outside points, the arrangement behaves as a point dipole with dipole moment $p = \rho \times 4/3\,\pi R^3 \times 2a = \frac{4}{3}\pi\sigma_0 R^3$

Differential Form Of Gauss Law For Spherically Symmetric Media:

Consider a region of space where charge density is a function of r alone. In such a case, Electric field magnitude is also a function of distance r due to radial symmetry. Now consider two Gaussian surfaces at radii r and $r+dr$ respectively. Let the corresponding field magnitudes be E and $E+dE$



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respectively. Applying Gauss law to the two surfaces:

$$(E+dE)\times 4\pi(r+dr)^2 = (q+dq)/\epsilon_0$$
$$-\quad (\quad E \times 4\pi r^2 \quad = q/\epsilon_0\ )$$
$$4\pi(E\times 2r\,dr + r^2 dE) = dq/\epsilon_0 \quad (\because \text{neglected second order terms})$$

Also, $dq = \rho \times 4\pi r^2 dr \Rightarrow 4\pi(2Er\,dr + r^2 dE) = \rho \times 4\pi r^2 dr/\epsilon_0$

dividing throughout by $4\pi r^2 dr$:

$$\boxed{\frac{dE}{dr} + \frac{2E}{r} = \rho/\epsilon_0}$$ ← Memorize this.

This equation now can be readily used for finding charge density function from the field function.

EXAMPLE: IRODOV 2.53

The field potential inside a charged ball depends only on the distance from its centre as $\phi = ar^2 + b$, where a and b are constants. Find the space charge distribution $\rho(r)$ inside the ball.

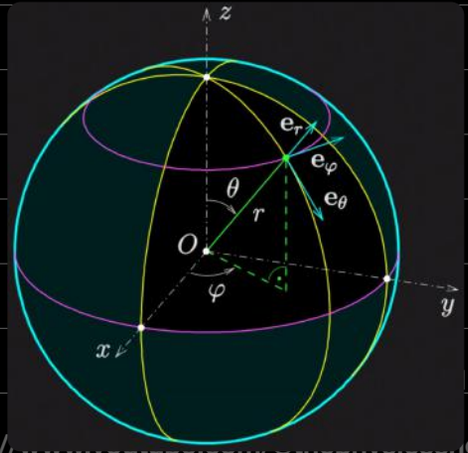
Solution: $E = -d\phi/dr = -2ar$

Now $\rho = \epsilon_0\left[\frac{dE}{dr} + \frac{2E}{r}\right] = \epsilon_0[-2a - 4a] = -6a\epsilon_0$ ans.

DIVERGENCE IN SPHERICAL COORDINATES: Recall that we had proved the differential form of Gauss law as $\vec{\nabla}\cdot\vec{E} = \rho/\epsilon_0$ and we had used cartesian coordinates for proving it. The same result also holds true in spherical and cylindrical coordinates, however the expression for divergence in spherical coordinates is complicated and here I am stating it without proof.

$$\vec{\nabla}\cdot\vec{E} = \frac{1}{r^2}\frac{\partial}{\partial r}[r^2 E_r] + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}[E_\theta \sin\theta] + \frac{1}{r\sin\theta}\frac{\partial}{\partial\phi}(E_\phi) = \rho/\epsilon_0$$

(No Need to Memorize this).



→ You can readily see that it reduces to our previous formula if $E_\theta = 0$ & $E_\phi = 0$.

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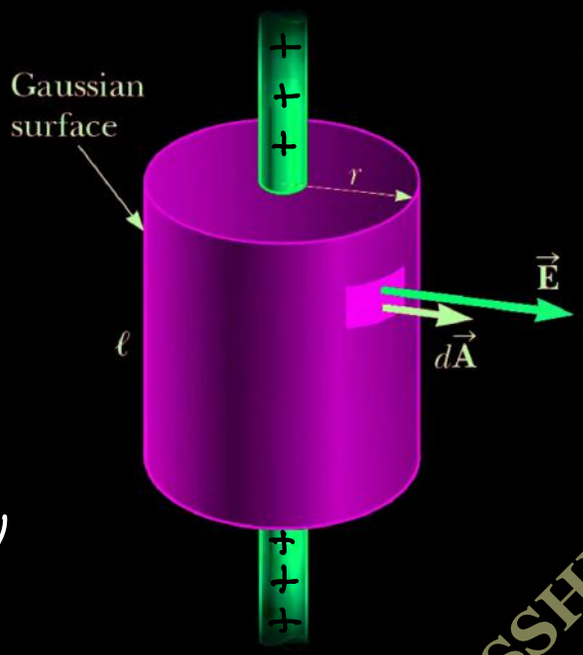
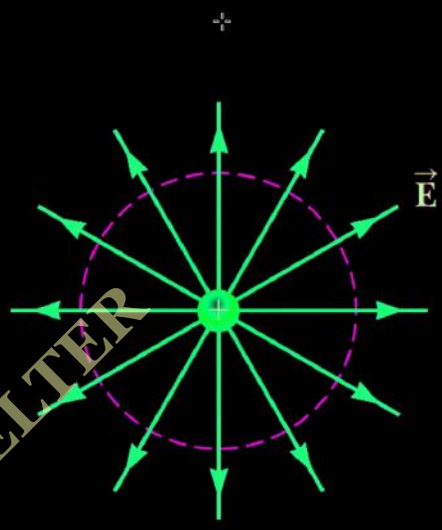
Gauss Law With Cylindrical Symmetry:

Field Due To An Infinite Wire:

Consider a Gaussian surface around an infinite wire having linear charge density λ as shown in the figure. Here the flux is associated only with curved surface of the cylinder due to up and down symmetry, field is parallel to flat surfaces of the Gaussian Cylinder.

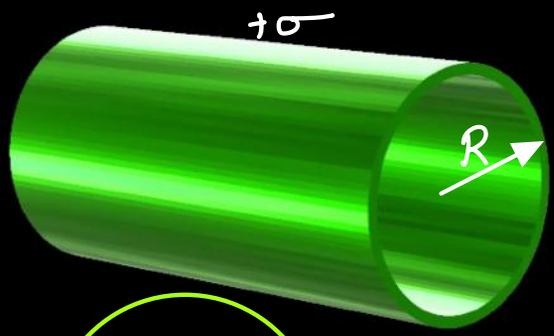




Now: $E \times 2\pi r \ell = \lambda \ell / \epsilon_0$ ($\because$ Gauss law)

$\Rightarrow$ $\boxed{E = \dfrac{\lambda}{2\pi\epsilon_0 r}}$

Field Due To An Infinfinite Cylindrical Shell

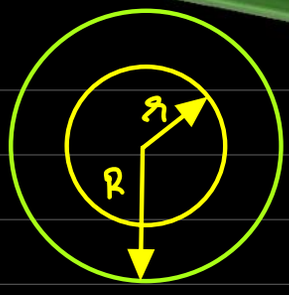


Here due to cylindrical symmetry, the field will be radially directed from axis of the cylinder. Now for $r<R$,

$E \times 2\pi r \ell = 0/\epsilon_0$ ($\because$ Gauss law)

$\Rightarrow$ $E = 0$

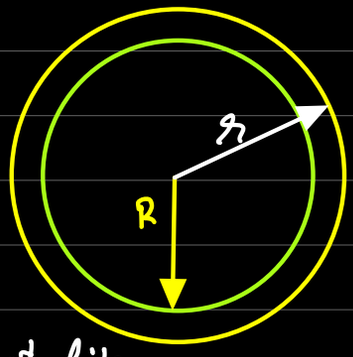
$\Rightarrow$ Just like a spherical shell an infinite cylindrical shell also produces no field at an inside point.



Now for $r>R$,

$$E \times 2\pi r \ell = \frac{\sigma \times 2\pi R \ell}{\epsilon_0}$$

$\Rightarrow$ $\boxed{E = \dfrac{\sigma}{\epsilon_0} \cdot \dfrac{R}{r}}$



Thus for outside points, field declines just like that of a line charge with $\lambda_{eq} = \sigma \times 2\pi R$.

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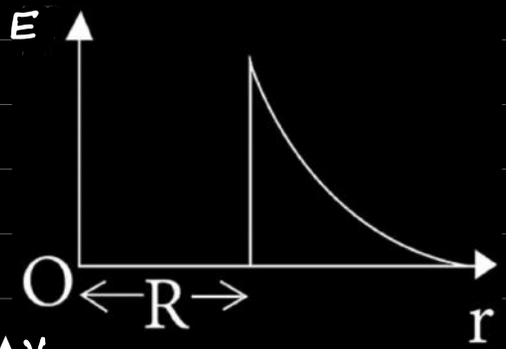
Field And Potential Graphs Of Thin Cylindrical Shell:

Now $E = -\frac{dV}{dr} = \frac{\sigma}{\epsilon_0}\frac{R}{r}$

$\Rightarrow$ $\boxed{V = \frac{\sigma R}{\epsilon_0}\ln\left(\frac{R}{r}\right)}$ (Assuming Zero potential at Surface).



Thus for outside points, the potential is also similar to that of a line charge and since field inside is zero, the potential is constant inside.

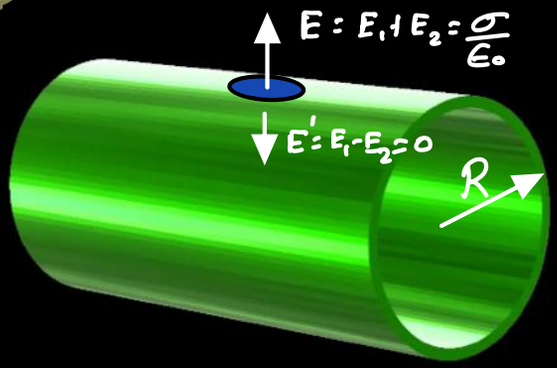
Electrostatic Pressure On Cylindrical Shell:

Consider a small patch on surface of a cylindrical shell, let the field due to local patch be E_1 and due to rest of the cylinder be E_2 then, just outside, $E = E_1 + E_2 = \sigma/\epsilon_0$ and just below the patch, $E_1 - E_2 = 0$

$\Rightarrow$ $E_1 = E_2 = \sigma/2\epsilon_0$. Now force on the patch, $dF = E_2 \times \sigma\, dS$ $\Rightarrow$ pressure $= \frac{dF}{dS} = \sigma E_2$

$\Rightarrow$

$\boxed{P = \sigma^2/2\epsilon_0}$



→ Now just as we did for spherical shell, we can find the force between two halves using the concept of pressure.

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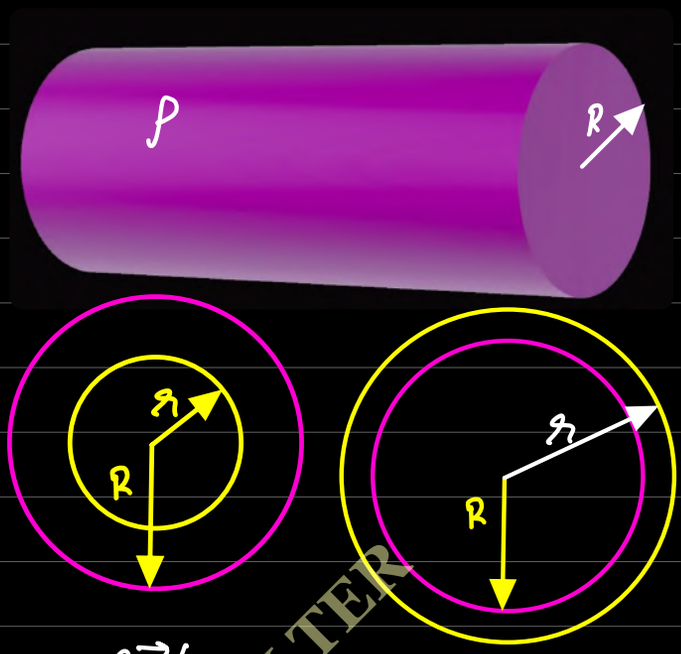
Field Due To A Solid Long Cylinder Of Charge:

Consider a solid long cylinder carrying a charge density ρ uniformly in its volume. Now applying Gauss law for $r < R$:

$$E \times 2\pi r l = \rho \times \pi r^2 l / \epsilon_0$$

$$\Rightarrow E = \frac{\rho r}{2\epsilon_0}$$, or vectorially:

$$\boxed{\vec{E} = \frac{\rho \vec{r}}{2\epsilon_0}}$$



constrast this with field inside a solid sphere which was $\rho\vec{r}/3\epsilon_0$.

→ By an argument similar to that for cavity in a solid sphere, for cylinder also, field in an axial cylindrical cavity can be shown to be constant (i.e $\rho\vec{a}/2\epsilon_0$) where $\vec{a}$ is $\perp$ vector between the axes of cavity and the charged cylinder).

Now for $r > R$, Applying Gauss Law:

$$E \times 2\pi r l = \rho \times \pi R^2 l / \epsilon_0 \Rightarrow \boxed{E = \frac{\rho R^2}{2\epsilon_0 r}}$$

This also just like the field of an infinite wire with $\lambda_{eff} = \rho\pi R^2$.
Also for potential, we can write: $E = -dV/dr$

$$\Rightarrow \boxed{V = \frac{\rho R^2}{2\epsilon_0} \ln(R/r)}$$

Assuming zero potential on the surface (for $r > R$).

For inside points, $E = \frac{\rho r}{2\epsilon_0} = -\frac{dV}{dr} \Rightarrow \boxed{V = \frac{\rho}{4\epsilon_0}[R^2 - r^2]}$

(assuming $V = 0$ at the surface).

For $r < R$.

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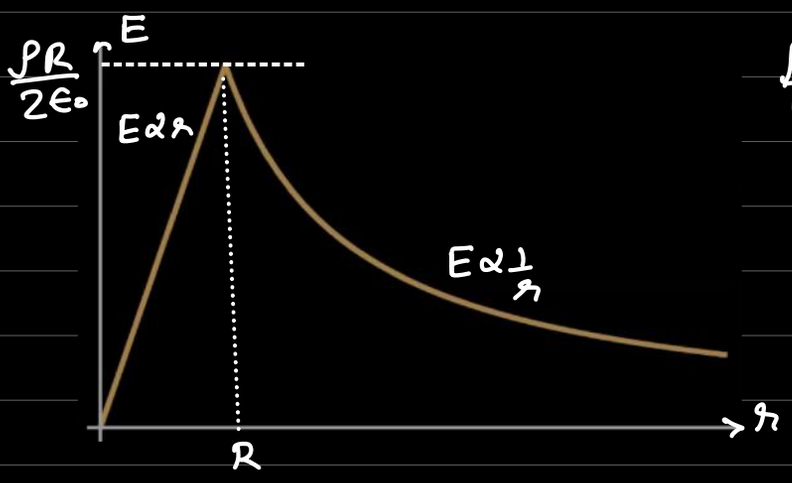
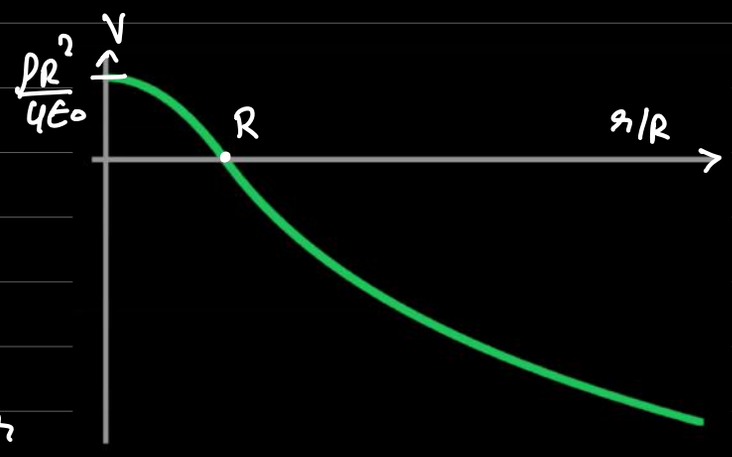
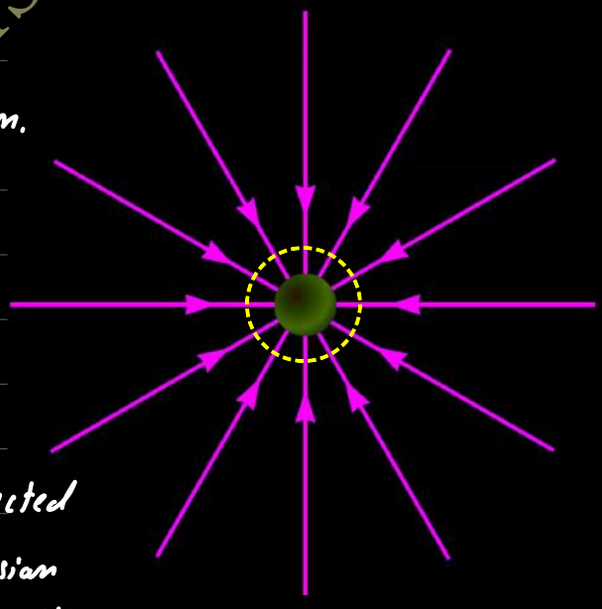
Field And Potential Graph For Solid Cylinder (Uniform Volume Charge):





Earnshaw's Theorem: Any point charge (or a collection of point charges) cannot be maintained in a stable stationary equilibrium configuration by the electrostatic interaction of charges alone.

Proof Of Earnshaw's Theorem:



Imagine a positive charge in stable equilibrium. For stability, if the charge is displaced in any direction, then it must experience a repulsion due to the universe back towards its stable position. Which means if we forget the positive point charges, all field lines of the universe must be directed towards this point. This means if a Gaussian surface is made around this point, there will be a negative flux associated with it. However after removal of the positive charge, there is nothing enclosed in this Gaussian Surface. This leads to violation of Gauss' Law. Thus, such kind of stable configuration is not possible for the positive point charge. (A similar argument can be built for stability of negative point charges).

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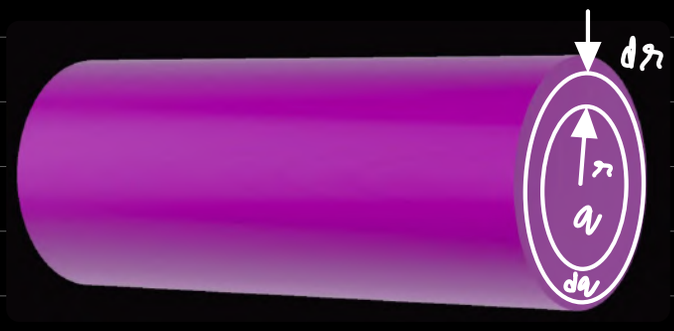
Differential Form Of Gauss Law In Cylindrical Symmetry:



Consider a Gaussian surface of radius r and another one of radius $r+dr$ with respective charges q and $q+dq$ enclosed within them. Now applying Gauss Law,

$$(E+dE)\times 2\pi(r+dr)l = (q+dq)/\epsilon_0$$

and $$-(E\times 2\pi r l = q/\epsilon_0)$$

$$2\pi l\,[E\,dr + r\,dE] = dq/\epsilon_0 = \rho \times 2\pi r l\,dr/\epsilon_0$$

Dividing throughout by $2\pi l\, r\, dr$:

$$\boxed{\frac{dE}{dr}+\frac{E}{r}=\rho/\epsilon_0}$$

Caution: This equation can be used only if E is a continuous and differentiable function of r.

Example: Irodov 3.24

The electric field strength depends only on the x and y coordinates according to the law $\mathbf{E} = a(x\mathbf{i}+y\mathbf{j})/(x^2+y^2)$, where a is a constant, $\mathbf{i}$ and $\mathbf{j}$ are the unit vectors of the x and y axis. Find the flux of the vector $\mathbf{E}$ through a sphere of radius R with its centre at the origin of coordinates.

Solution: $\vec{E} = \dfrac{a\,r\,\hat{r}}{r^2} = \dfrac{a}{r}\hat{r}$ (Recognise cylindrical Symmetry)

$$\frac{dE}{dr}+\frac{E}{r} = \frac{-a}{r^2}+\frac{a}{r^2} = 0 = \rho/\epsilon_0 \;!!!$$ (Field without any charge?!)

The problem here is that function is not continuous and differentiable at $r=0$. Recognise that field is just like that of a line charge where charge density is really zero everywhere in space except the point of discontinuity i.e $r=0$. For a line charge, $E = \lambda/2\pi\epsilon_0 r$.

By comparison, $\dfrac{\lambda}{2\pi\epsilon_0} = a \Rightarrow \lambda = 2\pi\epsilon_0 a$. Now the sphere intercepts a length $2R$ of this line $\therefore$ flux $= \dfrac{\lambda \times 2R}{\epsilon_0} = 4\pi R a$ ans.

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SELF ENERGY: Negative of work done by Electrostatic forces in assembling a continuous geometry of charges bringing its each infinitesimal charge from infinity is called the self energy of the geometry. Thus we could talk about self energy of a spherical shell, solid sphere, etc....

SELF ENERGY OF A UNIFORMLY CHARGED SPHERICAL SHELL:

Recall that negative of work done by electric field is also same as change in electrostatic potential energy. Imagine a thin shell of radius R on which a charge q is already accumulated and we are accumulating an additional dq.

Now at the surface, $V = \frac{kq}{R}$. Change in potential energy in bringing dq:

$$dU = V\,dq = \frac{kq}{R}\,dq$$, Thus self energy by the time a charge Q is accumulated:

$$U = \int_0^Q \frac{kq}{R}\,dq = \frac{kQ^2}{2R}$$

Thus $$\boxed{U = \frac{kQ^2}{2R} = \frac{1}{8\pi\epsilon_0}\frac{Q^2}{R}}$$

→ This self energy is stored in the space containing the electric field and we will get the same result if we integrate the energy Density $\frac{1}{2}\epsilon_0 E^2$ over the entire space.

SELF ENERGY OF A SHELL USING SPATIAL ENERGY DENSITY:

$$dU = \frac{1}{2}\epsilon_0 E^2\,dV, \quad E = \left(\frac{kQ}{r^2}\right), \quad dV = 4\pi r^2 dr$$

$$\Rightarrow dU = \frac{1}{2}\epsilon_0 \cdot \frac{1}{16\pi^2\epsilon_0^2}\frac{Q^2}{r^4} \times 4\pi r^2 dr$$

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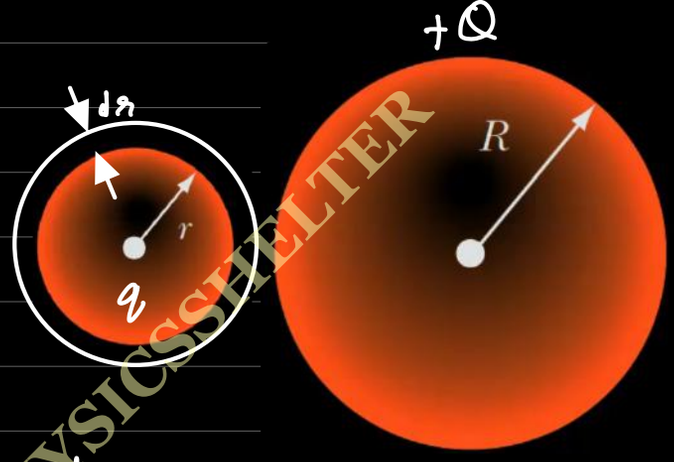
$$\Rightarrow dU = \frac{1}{8\pi\epsilon_0} Q^2 \int_R^\infty \frac{1}{r^2}\, dr = \frac{Q^2}{8\pi\epsilon_0 R}$$

This is same as the previous result. Hence we verified that self energy of a body is same as the energy associated with the electric field in entire space which is developed in the process of assembling the body bringing its elements from ∞.

Self Energy Of A Solid Sphere:



We wish to assemble a solid sphere having a total charge Q spread uniformly in a sphere of radius R. Suppose we have assembled up to radius r and assembling a further radius dr, then, potential of sphere of radius r: $V = \frac{kq}{r}$, Now when we bring an additional dq, $dU = V\,dq = \frac{kq}{r}\,dq$

Also $q = \frac{Q}{\frac{4}{3}\pi R^3} \cdot \frac{4}{3}\pi r^3 = \frac{Q r^3}{R^3} \Rightarrow r = R\left(\frac{q}{Q}\right)^{1/3}$

Thus $dU = \frac{kq}{R}\left(\frac{Q}{q}\right)^{1/3} dq = \frac{kQ^{1/3}}{R}\, q^{2/3} dq$

$$\Rightarrow U = \frac{kQ^{1/3}}{R}\int_0^Q q^{2/3}\, dq = \frac{3}{5}\frac{kQ^{1/3}}{R}\left[Q^{5/3} - 0\right] = \frac{3}{5}\frac{kQ^2}{R}$$

Thus for solid sphere: $\boxed{U = \frac{3}{5}\frac{kQ^2}{R}}$

Self Energy Of Solid Sphere By Integrating Field Energy:

Recall that field Energy outside a spherical shell was $\frac{kQ^2}{2R}$, so this part will be same due to solid sphere as well.

We only need to calculate the additional field energy stored in the volume of the sphere.

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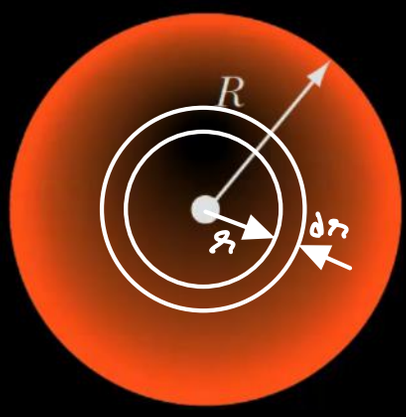
Now field inside the sphere: $E = \frac{kQr}{R^3}$



$dV = 4\pi r^2 dr$, $\Rightarrow dU = \frac{1}{2}\epsilon_0 E^2 dV$

$= \frac{1}{2}\epsilon_0 \frac{1}{16\pi^2\epsilon_0^2} \cdot \frac{Q^2 r^2}{R^6} \times 4\pi r^2 dr$

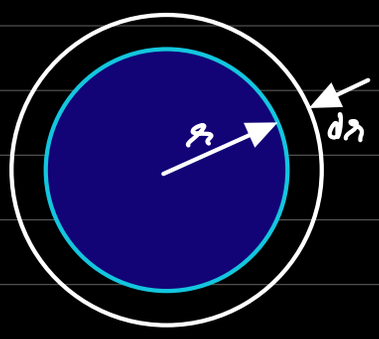
$= \frac{1}{8\pi\epsilon_0} \frac{Q^2}{R^6} r^4 dr \Rightarrow U = \frac{Q^2}{8\pi\epsilon_0 R^6} \int_0^R r^4 dr$

$= \frac{Q^2}{40\pi\epsilon_0 R^2} = \frac{1}{10}\frac{kQ^2}{R}$, Thus total $U = U_{in} + U_{out}$

$= \frac{kQ^2}{2R} + \frac{kQ^2}{10R} = \frac{3}{5}\frac{kQ^2}{R}$, This is the same result as we obtained above. Once again we have verified that self energy is same as the field energy stored in space.

Self Energy Of A Uniformly Charged Disc :

We wish to assemble a uniform disc of radius R and charge density σ. Suppose we have already assembled up to radius r and further adding a radius dr. Recall that potential at periphery of such a disc is given as $V = \frac{\sigma r}{\pi\epsilon_0}$



Thus $dU = V dq = \frac{\sigma r}{\pi\epsilon_0} \cdot \sigma \times 2\pi r dr = \frac{2\sigma^2}{\epsilon_0} r^2 dr$

Thus total self energy $U = \frac{2\sigma^2}{\epsilon_0} \int_0^R r^2 dr = \frac{2}{3}\frac{\sigma^2}{\epsilon_0} R^3$ Ans.

also $\sigma = Q/\pi R^2$ and $\epsilon_0 = 1/4\pi k$ $\Rightarrow$ $\boxed{U = \frac{8kQ^2}{3\pi R}}$

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Total Electrostatic Energy Of A System: The sum of all the interaction energies and self energies of a system is called the total electrostatic energy of a system. (Note that for point charges, self energy is infinite however we usually take it as zero, which is ok. so long as point charges are intact (not being split) ∵ we are mostly interested in changes of potential energies and constant terms do not matter so long as they come as it is on LHS and RHS of the equation).

→ For shells and solid spheres, interaction energies are just like those of point charges

Example: A solid sphere and a spherical shell each having a charge Q are just touching each other as shown in the figure:

(a) Write total Electrostatic Energy
(b) What is the work done by an external agent in slowly shrinking the spheres to half of their initial radii keeping their centres locations intact
(c) What is the work done by an external agent in slowly doubling their centre to centre separation keeping their shapes intact?

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Solution: (a) Total self Energy $U_s = \frac{kQ^2}{2R} + \frac{3kQ^2}{5R} = \frac{11}{10}\frac{kQ^2}{R}$

Total Interaction Energy: $U_I = \frac{kQ^2}{2R}$ (∵ centre to centre is 2R).

∴ Total Electrostatic Energy $U = U_s + U_I = \frac{8kQ^2}{5R}$

(b) If spheres shrink while maintaining their centres, only self energies change, final self Energy, $U_{sf} = \frac{kQ^2}{2(R/2)} + \frac{3kQ^2}{5(R/2)}$

$\Rightarrow U_{sf} = \frac{11}{5}\frac{kQ^2}{R}$, Now $\Delta W_{ext} = \Delta U = \left(\frac{11}{5} - \frac{11}{10}\right)\frac{kQ^2}{R}$

$\Rightarrow \quad \Delta W_{ext} = \frac{11}{10}\frac{kQ^2}{R}$ Ans.

(c) If spheres are moved apart maintaining their shapes, only interaction energy changes, $U_{I_f} = \frac{kQ^2}{2(2R)} = \frac{kQ^2}{4R}$

Now work done by external agent $= \Delta W_{ext} = \Delta U = \frac{kQ^2}{4R} - \frac{kQ^2}{2R}$

$\Rightarrow \quad \Delta W_{ext} = -\frac{kQ^2}{4R}$ Ans.

→ If both phenomena were to occur simultaneously, the required work will be change in total electrostatic energy which is simply the superposition of the two works.

→ Self energy can be thought of as energy required for assembling the geometries remote from each other and interaction Energy can be thought of as work done in bringing them close together.

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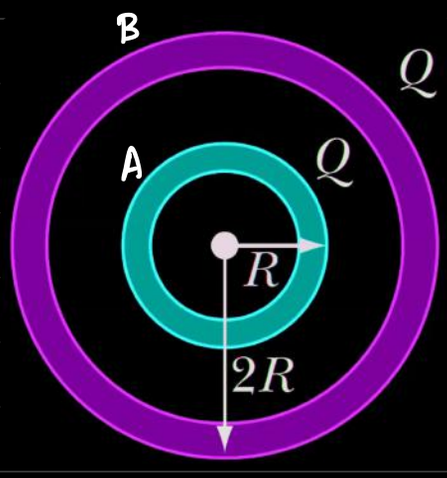
EXAMPLE: Find the total Electrostatic Energy of the system of two uniformly charged thin shells as shown in the figure.



Solution: Total electrostatic energy is the sum of self energies and the energy required to bring them in to final relative configuration. i.e. Work done against own repulsions and work done against repulsion of each other. Here self energy part is simply calculated as:

$$U_S = U_A + U_B = \frac{kQ^2}{2R} + \frac{kQ^2}{2(2R)} = \frac{3kQ^2}{4R}$$

Now interaction energy part can be calculated using two visualisations:
(I) Keeping A fixed arranging particles of B around A.
(II) Keeping B fixed inserting shell A inside B.

(I) Now potential at distance $2R$ due to A = $V_{BA} = \frac{kQ}{2R}$ ($\because$ outside point)

$\Rightarrow U_{AB} = Q_B \times V_{AB} = \frac{kQ^2}{2R} \Rightarrow U_{AB} = \frac{kQ^2}{2R}$

(II) If shell A is inserted in B, work is done only till bringing its particles till surface, after that no work is done because inside of B all volume is equipotential due to itself.

Now potential at surface of B:

$$V_{BB} = \frac{kQ}{2R} \quad (\because \text{Surface point})$$

Thus $U_{AB} = Q_A \times V_{BB} = \frac{kQ^2}{2R}$, this is the same interaction energy that we obtained previously using the first visualisation.

Now Total Electrostatic Energy $= U_s + U_{AB} = \frac{3}{4}\frac{kQ^2}{R} + \frac{kQ^2}{2R}$

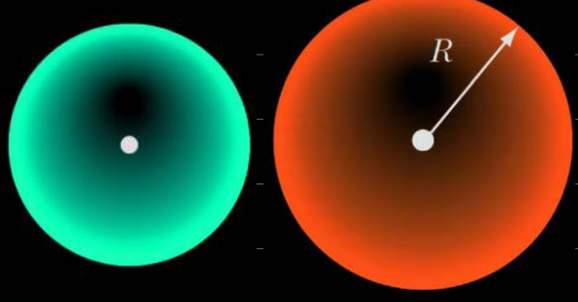
$\Rightarrow U = \frac{5kQ^2}{4R}$ Ans.

→ Even if shells were not perfectly concentric, Total Electrostatic Energy (and interaction energy) will be still the same so long as A is completely inside B ($\because$ By shell theorem, moving A within B

→ Alternatively we could find the electrostatic energy of the system using $U = \int \frac{1}{2}\epsilon_0 E^2 d\forall$.

Interaction Energy Of Two Bodies As $\epsilon_0 \int \vec{E_1} \cdot \vec{E_2} \, d\forall$:

Recall that electrostatic energy is same as the energy associated with electric field i.e $\int \frac{1}{2}\epsilon_0 E^2 d\forall$.



Now suppose we have two bodies producing electric fields $\vec{E_1}$ and $\vec{E_2}$, then electrostatic energy associated with the system must be

$$U = \int \frac{1}{2}\epsilon_0 |\vec{E_1} + \vec{E_2}|^2 d\forall$$

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$$\Rightarrow U = \underbrace{\int \frac{1}{2}\epsilon_0 E_1^2 \, d\tau}_{\text{(I)}} + \underbrace{\int \frac{1}{2}\epsilon_0 E_2^2 \, d\tau}_{\text{(II)}} + \underbrace{\int \epsilon_0 E_1 E_2 \cos\theta \, d\tau}_{\text{(III)}}$$

Now recognise that terms (I) and (II) represent respectin self energies, therefor the thind term must represent the interaction energy of the two spheres.

Thus $\boxed{U_{12} = \int \epsilon_0 (\vec{E}_1 \cdot \vec{E}_2) \, d\tau}$

EXAMPLE: MORIN & PURCELL (1.8) :

Oscillating in a ring ***

A ring with radius R has uniform positive charge density λ. A particle with positive charge q and mass m is initially located at the center of the ring and is then given a tiny kick. If it is constrained to move in the plane of the ring, show that it undergoes simple harmonic motion (for small oscillations), and find the frequency. *Hint:* Find the potential energy of the particle when it is at a (small) radius, r, by integrating over the ring, and then take the negative derivative to find the force. You will need to use the law of cosines and also the Taylor series $1/\sqrt{1+\epsilon} \approx 1 - \epsilon/2 + 3\epsilon^2/8$.

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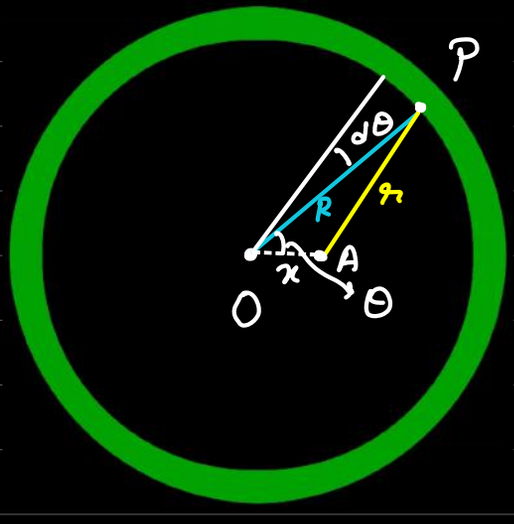
Solution: Suppose particle is displaced through a small distance x in the plane of the ring as shown. We shall find the force on the particle by finding the potential and the field by differentiating the potential. Consider an element of the ring at an angle θ and angular thickness $d\theta$ as shown. Using Cosine law, distance of P from A is $\sqrt{R^2+x^2-2Rx\cos\theta}$



$$\Rightarrow dV = \frac{k\lambda R d\theta}{\sqrt{R^2+x^2-2Rx\cos\theta}} = k\lambda \, d\theta \left[1 + \left(\frac{x}{R}\right)^2 - 2\frac{x}{R}\cos\theta\right]^{-1/2}$$

$$\Rightarrow dV \approx k\lambda d\theta \left[1 - \frac{1}{2}\left(\left(\frac{x}{R}\right)^2 - \left(\frac{2x}{R}\right)\cos\theta\right) + \frac{3}{8}\left(\left(\frac{x}{R}\right)^2 - \left(\frac{2x}{R}\cos\theta\right)\right)^2\right]$$

Now we need to integrate only even powers of $\cos\theta$ because from 0 to 2π integration of all odd powers of $\cos\theta$ will be zero. Also for field function we are interested only in up to coefficients of x^2. (Dropping x^4 terms and odd powers of $\cos\theta$):

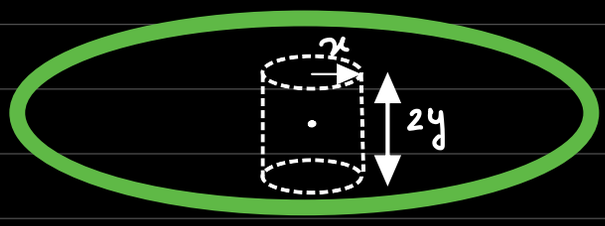
Thus $$E = -\frac{d}{dx}\int_0^{2\pi} k\lambda d\theta \left[1 - \frac{1}{2}\frac{x^2}{R^2} + \frac{3}{2}\frac{x^2}{R^2}\cos^2\theta\right]$$

$$\Rightarrow E = -\frac{d}{dx}\left[k\lambda\left[2\pi + \frac{\pi x^2}{2R^2}\right]\right) = -\frac{k\lambda\pi x}{R^2} = \frac{-\lambda x}{4\epsilon_0 R^2}$$

$$\Rightarrow a = \frac{Eq}{m} \Rightarrow \frac{d^2x}{dt^2} = \frac{-\lambda q}{4\pi m\epsilon_0 R^2}x \Rightarrow \omega = \frac{1}{R}\sqrt{\frac{\lambda q}{4\pi m\epsilon_0}}$$

or $$\omega = \sqrt{\frac{Qq}{8\pi\epsilon_0 m R^3}}$$ ans.

Aliter: Imagine a small pill shaped gaussian surface around the centre of the ring with radius x and a height $2y$. Net flux through this cylinder must be zero. Let Radial field associated with its curved surface be $E(x)$, Axial field associated with flat surface is $\frac{kQy}{(R^2+y^2)^{3/2}}$



Now applying Gauss law: $-E \times 2\pi x \times 2y + \frac{kQy}{(R^2+y^2)^{3/2}} \times \pi x^2 \times 2 = 0$

$$\Rightarrow E = \frac{kQx}{2(R^2+y^2)^{3/2}} \approx \frac{kQx}{2R^3} \quad (\because y << R)$$

Now $$m\frac{d^2x}{dt^2} = \frac{-kQqx}{2R^3} \Rightarrow \omega = \sqrt{\frac{Qq}{8\pi\epsilon_0 m R^3}}$$

This is same result as obtained above.

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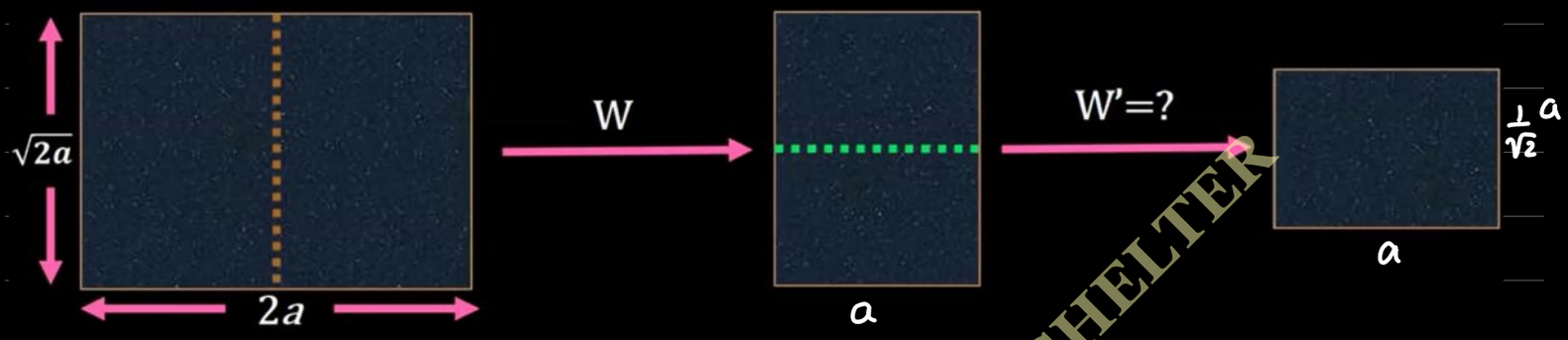
EXAMPLE: (AMBARISH ORIGINAL):

A Beautiful Origami of Electrostatics

Challenge: *A uniformly charged very thin sheet of width $2a$ and height $\sqrt{2}a$ is slowly folded about a vertical symmetry axis as shown. The work required in process is W. How much work is required to fold it further about a horizontal axis?*



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Solution: Here we do not know the exact formula for self energy, however we know that it should be proportional to Q^2/Linear Dimension. (How? All the results are like that, Shell: $kQ^2/2R$, Solid Sphere $3kQ^2/5R$, Uniform Disc $\frac{8}{3\pi}\frac{Q^2}{R}$) etc. So let formula for self energy for sheet be:

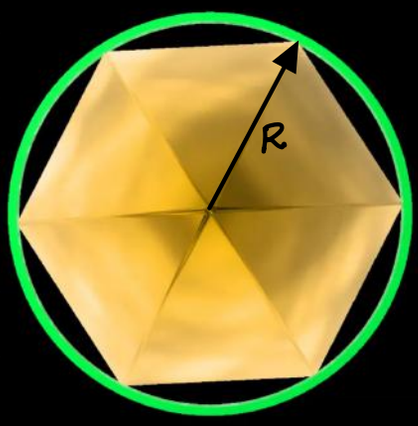
$$U = CQ^2/\ell \quad \text{where } \ell \text{ is the longer side.}$$

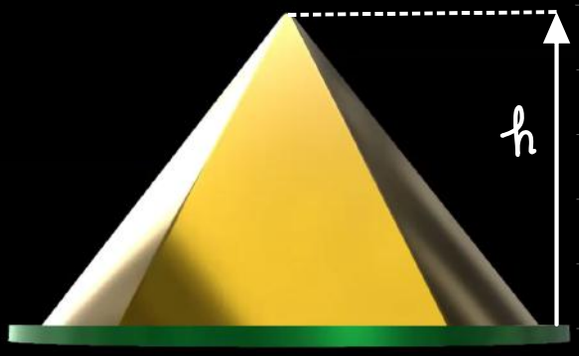
Thus $W = \left(\frac{CQ^2}{\sqrt{2}a} - \frac{CQ^2}{2a}\right)$, $W' = \frac{CQ^2}{a} - \frac{CQ^2}{\sqrt{2}a}$

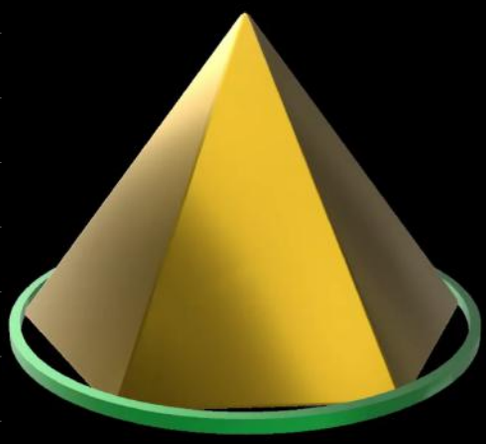
$$\Rightarrow \frac{W'}{W} = \left(\frac{1}{a} - \frac{1}{\sqrt{2}a}\right) \Big/ \left(\frac{1}{\sqrt{2}a} - \frac{1}{2a}\right) = \left(\frac{2-\sqrt{2}}{\sqrt{2}-1}\right) = \sqrt{2}$$

$$\Rightarrow W' = \sqrt{2}W \text{ Ans.}$$

Solid Angle Of n-Gonal Pyramid.







In general spherical trigonometry and solid geometry is not much expected in JEE However one question was asked in JEE main to which wrong key was given . The correct solution requires the result of solid angle of a regular pyramid. Here it is stated without proof:

$$\Omega = 2\pi - 2n \tan^{-1}\left(\frac{\tan \pi/n}{\sqrt{1+r^2/h^2}}\right)$$

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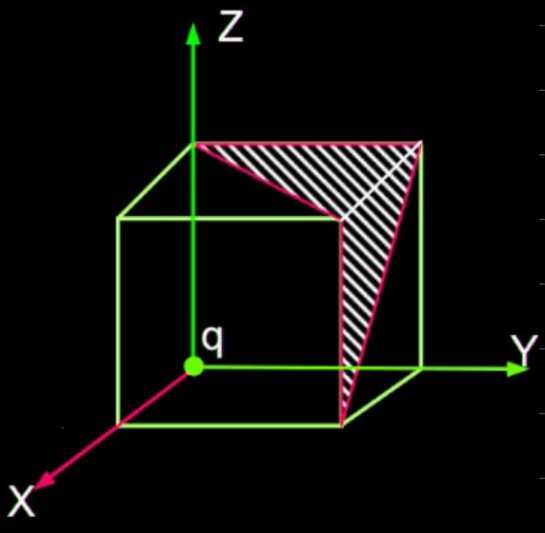
Example: (JEE Main 2021):

A charge 'q' is placed at one corner of a cube as shown in figure. The flux of electrostatic field $\vec{E}$ through the shaded area is :



(A) $\frac{q}{8\varepsilon_0}$

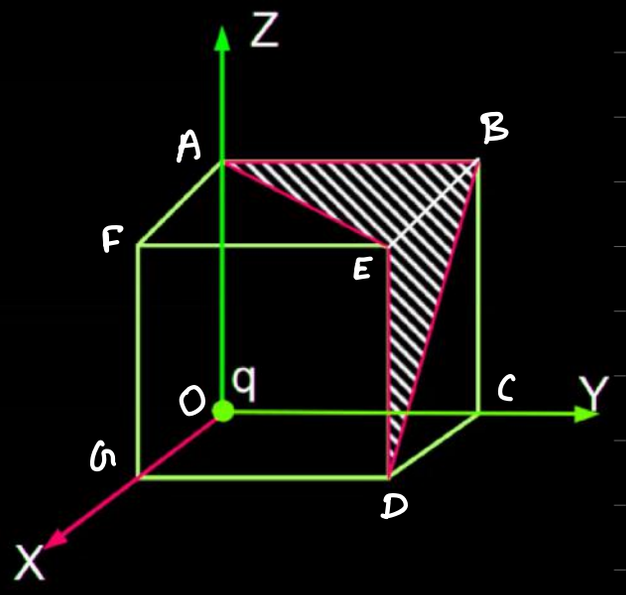
(B) $\frac{q}{4\varepsilon_0}$

(C) $\frac{q}{24\varepsilon_0}$

(D) $\frac{q}{48\varepsilon_0}$

<u>Solution:</u> We have already solved a similar problem where we know that $\phi_{BCDE} = \phi_{ABEF} = q/24\epsilon_0$, while by symmetry we can say that $\phi_{ABE} = \phi_{AFE}$

$= q/48\epsilon_0$ however we can't equate the flumes of BCD and BED because symmetry is broken. BCD is "nearer" to the charge than BED.

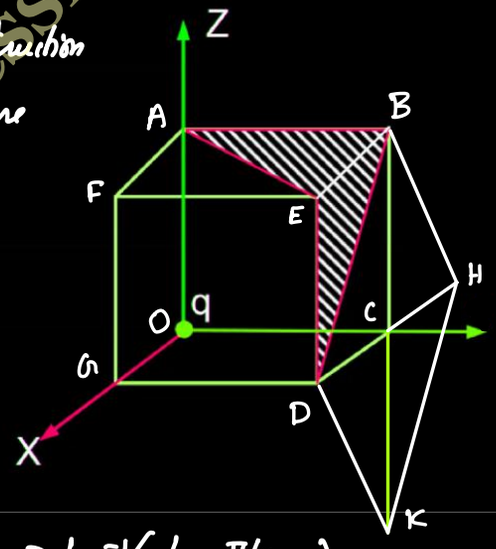


So to solve the problem, our idea is to subtract the flum of BCD from $q/24\epsilon_0$. It turns BCD can be modeled as a symmetric portion of a regular pyramid.

In the adjacent figure we have shown by construction the square BDKH which forms a regular square base pyramid at O. Now let us imagine each side of cube to be one unit.

$\therefore$ For pyramid BDKH, altitude is $h = 1$ and circumradius, $R = 1$ (i.e CD).

Thus solid angle of the pyramid:



$$\omega = 2\pi - 2n\,\text{Tan}^{-1}\left(\frac{\tan\pi/n}{\sqrt{1+R^2/h^2}}\right) = 2\pi - 8\tan^{-1}\left(\frac{\tan\pi/4}{\sqrt{2}}\right)$$

$$\omega_{BDKH} = 2\pi - 8\tan^{-1}(1/\sqrt{2}) \Rightarrow \omega_{BCD} = \frac{1}{4}\omega_{BDKH}$$

$$\Rightarrow \omega_{BCD} = \frac{\pi}{2} - 2\tan^{-1}(1/\sqrt{2}), \quad \text{Now } \phi = \frac{\omega}{4\pi} \times q/\epsilon_0$$

$$\Rightarrow \phi_{BCD} = \left[\frac{1}{8} - \frac{1}{2\pi}\tan^{-1}(1/\sqrt{2})\right] q/\epsilon_0$$

$$\Rightarrow \phi_{BDE} = \frac{q}{24\epsilon_0} - \phi_{BCD} = \frac{q}{\epsilon_0}\left[\frac{1}{2\pi}\tan^{-1}(1/\sqrt{2}) - \frac{1}{12}\right]$$

Thus required net flum: $\phi_{ABE} + \phi_{BDE} = \frac{q}{48\epsilon_0} + \phi_{BDE} = \frac{q}{\epsilon_0}\left[\frac{1}{2\pi}\tan^{-1}\left(\frac{1}{\sqrt{2}}\right) - \frac{3}{48}\right]$ Ans

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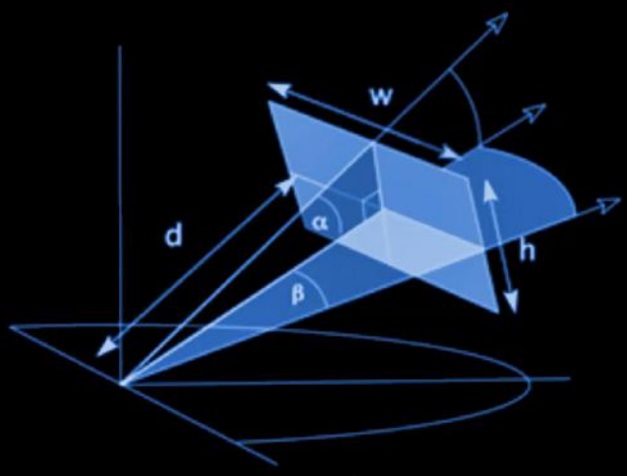
Note alternatively we could use formula for solid angle of a rectangle given by:

$$\omega = 4\cos^{-1}\frac{\sqrt{1+\tan^2\alpha+\tan^2\beta}}{\sec\alpha\sec\beta}$$



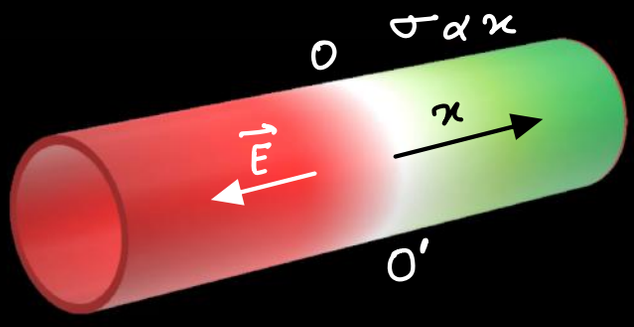
(See adjacent figure)

(Students are not required to memorize these solid angle formulae).

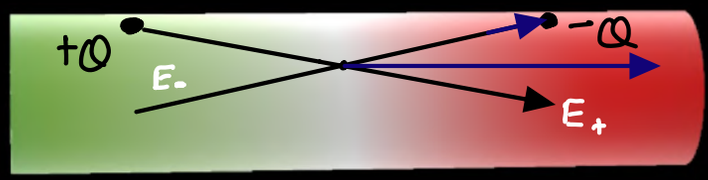
EXAMPLE: BUKHOVTSEV 429:

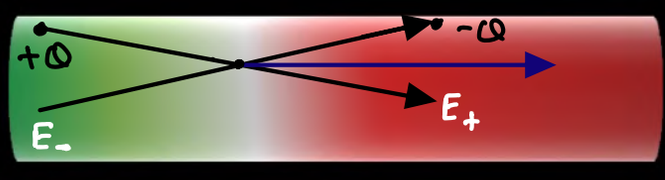
429. Charges are so arranged on the surface of an infinitely long cylinder that the right-hand half of the cylinder surface from section OO' is charged positively, and the left-hand half negatively (Fig. 152). The density of the charges increases in both directions directly proportional to the distance from section OO'.

Prove that the intensity of the electric field at all points inside the cylinder will be the same everywhere and directed along the axis of the cylinder as shown in the figure by the arrow.



Solution: Clearly at all points of the mid section, the field must be horizontal because for every positive charge, there exists an equally "beautiful" antisymmetric negative charge. Now suppose we superpose a uniform negative charge density over and above the initial distribution, it should not change the field anywhere ∴ field inside a uniform shell is zero. However this shift the neutral section leftward and maintains a linear distribution of $\sigma \propto x$ wrt this new neutral point. Thus we can say that field across this new neutral section is also horizontal everywhere.





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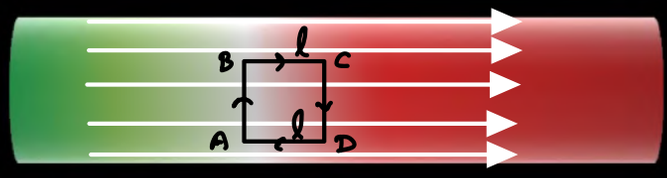
However in the process of superposing a uniform negative charge, we did not change the field anywhere, therefore our field must have been horizontal in the first place itself. Thus we have proved that field inside this pipe is throughout horizontal.

Now consider a closed loop ABCDA as shown, net work done along this closed loop must be zero, because electrostatic field is a conservative field.



Now $W_{AB} = W_{CD} = 0$ ($\because$ moving $\perp$ to field lines)

$$\therefore \quad \vec{E}_{BC} \cdot \vec{BC} + \vec{E}_{AD} \cdot \vec{DA} = 0 \Rightarrow E_{BC}\, l - E_{AD}\, l = 0$$

$\therefore \quad E_{BC} = E_{AD} = E$ (say). Thus the electric field is uniform throughout such a pipe. (QED).

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